Mini Project 1

Two Stage OTA Design

Part 1: gm/ID Design Charts

NMOS Charts:

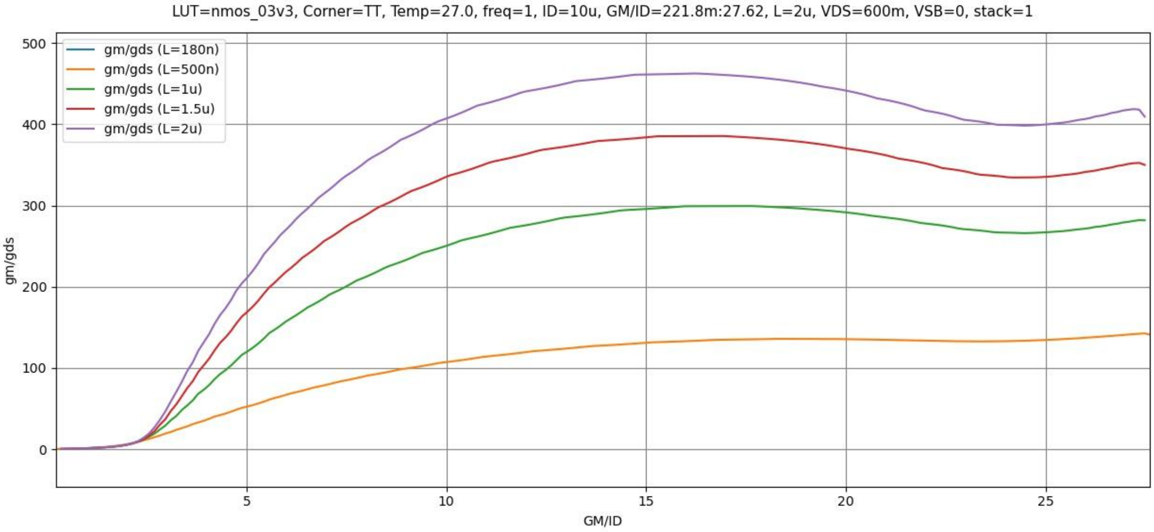


Figure 1 gm/gds vs gm/id

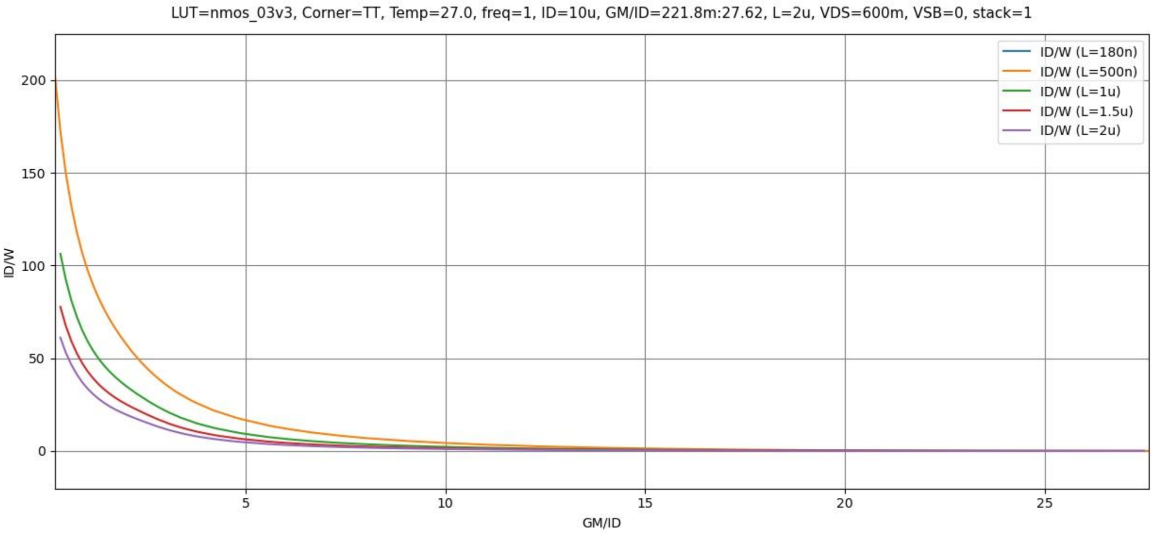


Figure 2 ID/W vs gm/ID

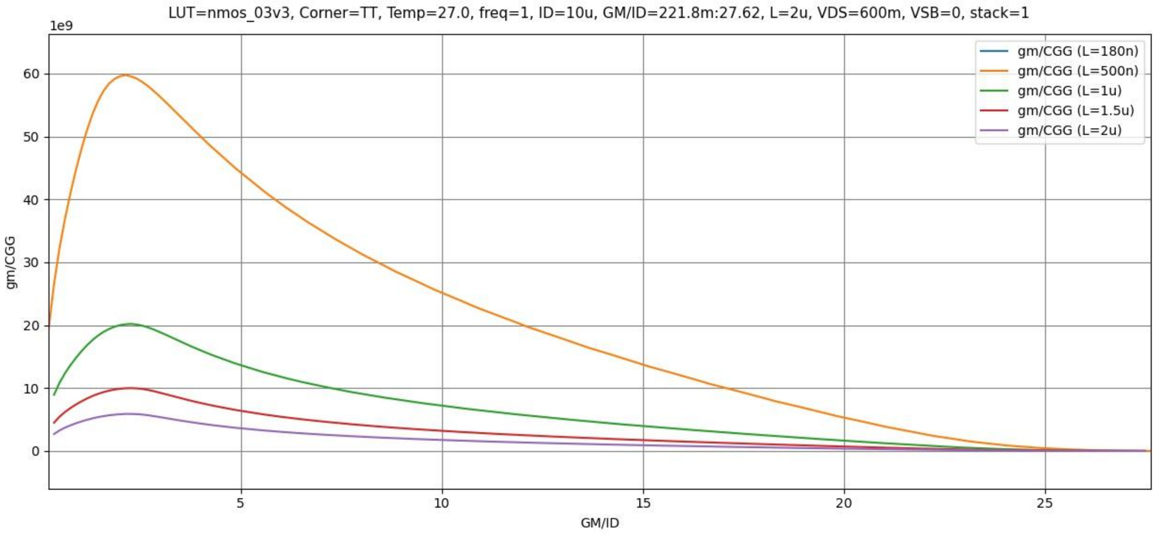


Figure 3 gm/Cgg vs gm/ID

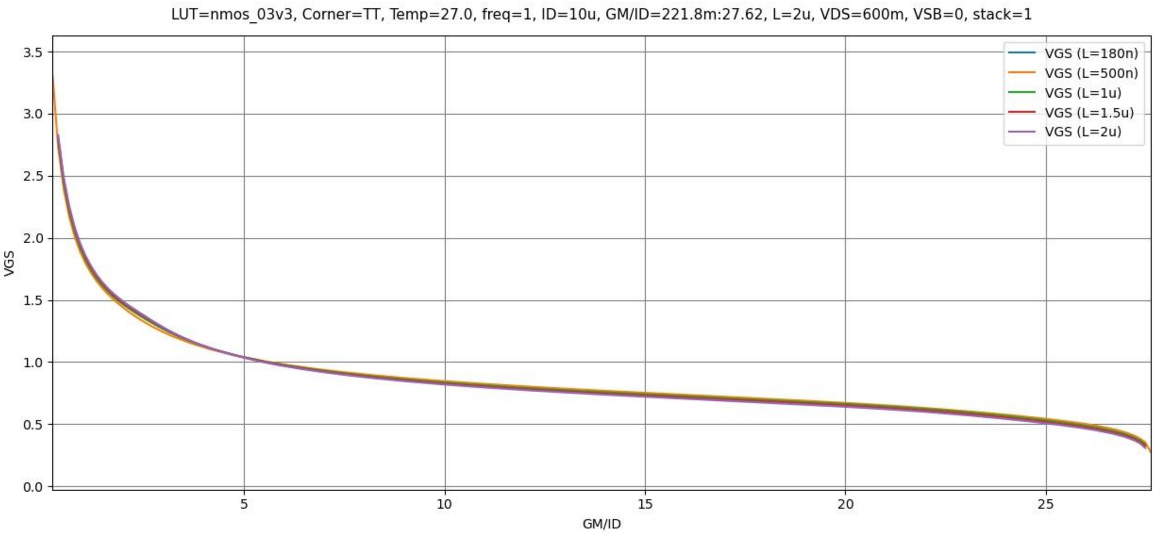


Figure 4 VGS vs gm/ID

PMOS Charts:

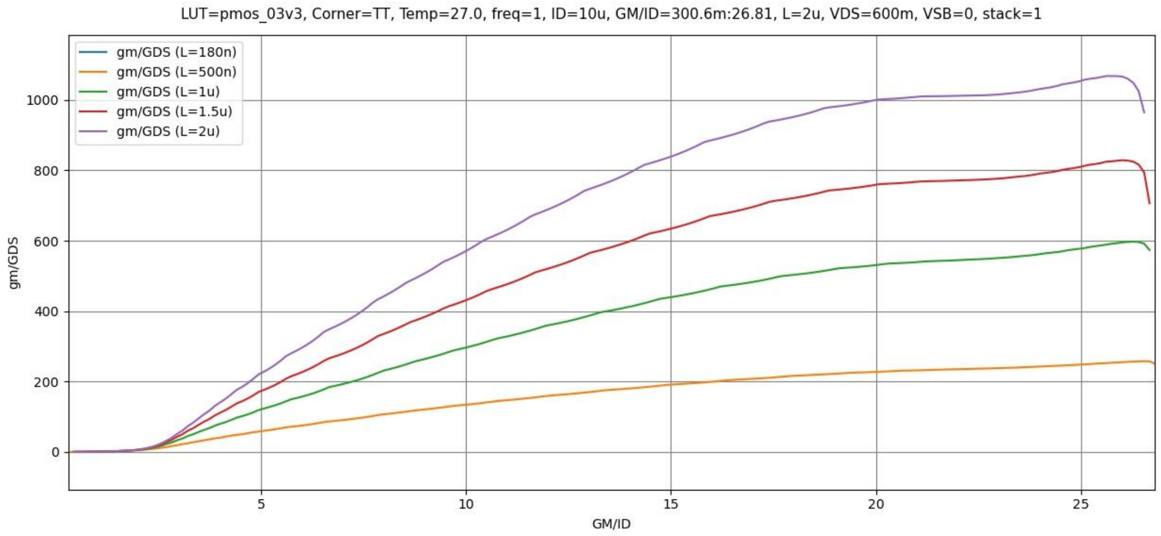


Figure 5 gm/gds vs gm/ID

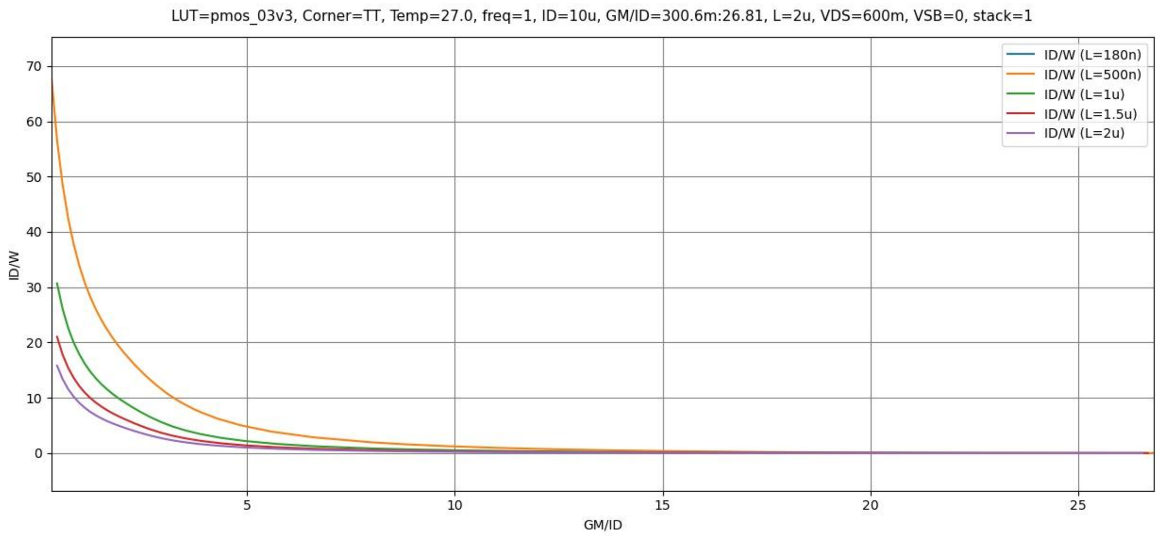


Figure 6 ID/W vs gm/ID

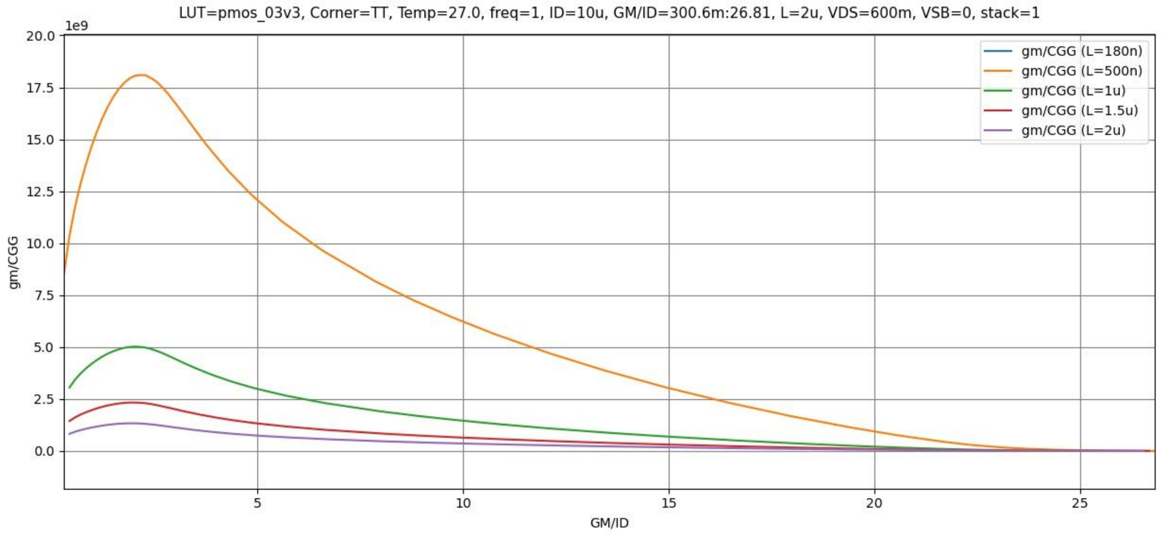


Figure 7 gm/Cgg vs gm/ID

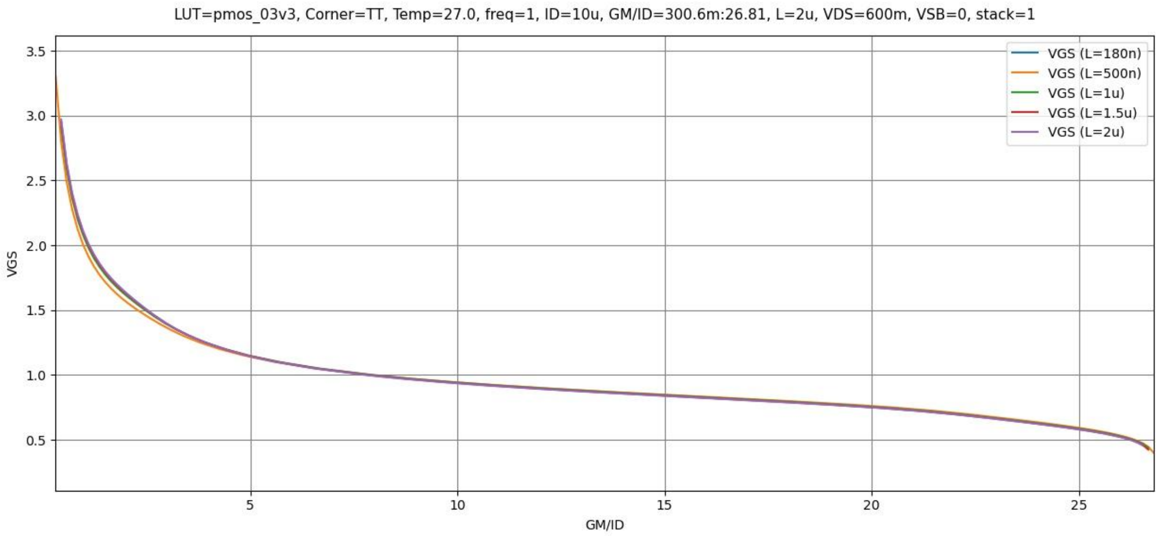


Figure 8 VGS vs gm/ID

Part 2: OTA Design

Required Specifications:

Technology	0.18um
Supply voltage	1.8V
Static gain error	<= 0.05%
CMRR @ DC	>= 74dB
Phase margin (avoid pole-zero doublets)	>= 70°
OTA current consumption	<= 60uA
CMIR – high	>= 1V
CMIR – low	<= 0.2V
Output swing	0.2 – 1.6V
Load	5pF
Buffer closed loop rise time (10% to 90%)	<= 70ns
Slew rate (SR)	$5V/\mu s$

Design Observations:

$$Static\ Gain\ Error = \frac{1}{LG} = \frac{1}{A_{OL}\beta} \because \beta = 1 \therefore \frac{1}{A_{OL}} = \frac{0.05}{100} \rightarrow \boldsymbol{A_{OL} = 2000}$$

- The Open Loop Gain required is large and should be achieved by a two stage OTA
- The CMIR required is close to the Ground Rail, thus a PMOS input stage is preferred.
- The output swing is quite large, thus a CS amplifier with a current source load is preferred for the output stage
- The requirement for the differential input would be suited better by a 5T-OTA.

A Miller-OTA topology will be used
It is required to size about 4 pairs of MOSFETS.

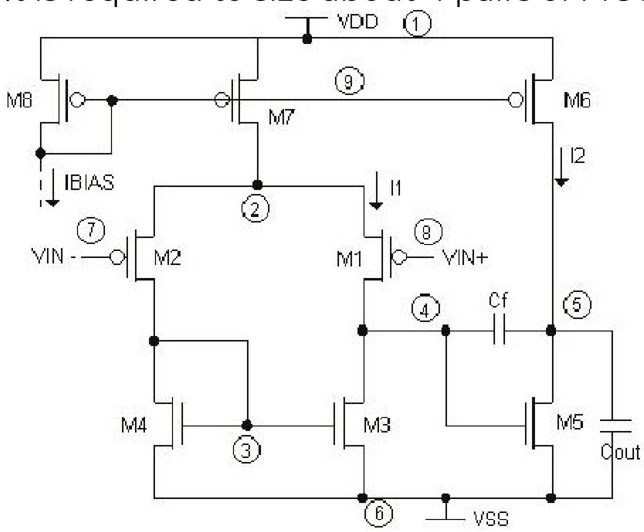


Figure 9 Miller OTA Topology

Miller Capacitance:

A reasonable choice for Miller Capacitance is $0.5 * C_L$

$$C_C = 0.5 * 5p = \mathbf{2.5pF}$$

Input Transistors:

$$t_{rise} = 70ns = 2.2\tau \rightarrow \tau = 31.82ns = \frac{1}{\omega_u}$$

$$\omega_u = 31.427Mrad = \frac{gm_{1,2}}{C_C} \rightarrow \boldsymbol{gm_{1,2} = 78.567\mu S \approx 80\mu S}$$

Open Loop Gain can be divided between the two stages:

$$A_{OL} \geq 66dB = 36\ (5T\ OTA\ Gain) + 30\ (CS\ Gain)$$

$$63.1 \approx \frac{gm_{1,2}ro}{2} (Assuming\ Both\ NMOS\ and\ PMOS\ have\ the\ same\ gds)$$

$$\rightarrow \boldsymbol{ro = 1.5774M\Omega \approx 1.6M\Omega}$$

Using the spec of Slew Rate:

$$SR = \frac{I_{B1}}{C_C} = \frac{5V}{\mu S} \rightarrow \boldsymbol{I_{B1} = 12.5\mu A \rightarrow I_{D1,2} = 6.25\mu A}$$

Assuming the Voltage Drop is divided equally across all 3 transistors in branch, VDS of each transistor = 0.6V

Using these three parameters we can get the width and length of the input pair quickly using ADT:

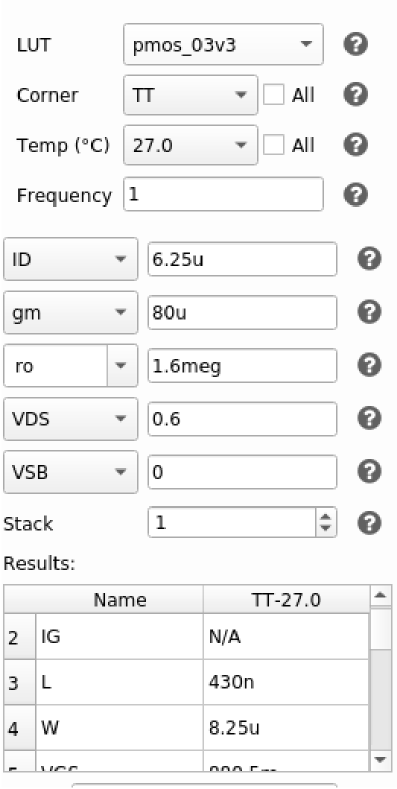


Figure 10 Intial Values for the input pair from SA

$$\boldsymbol{L = 430nm \quad , \quad W = 8.25\mu m}$$

These are acceptable values for the Input Pair and decent for initial design!

PMOS Current Mirror Load:

Using CMIR-Max:

$$CMIR_{LOW} = 0.2 = -Vth_{1,2} + Vgs_{3,4}\ (Using\ Value\ of\ Vth_{1,2}\ from\ SA)$$

$$\rightarrow Vgs_{3,4} = 0.98 \approx 1V$$

The value for VGS is slightly large so I'll use a lower value of 0.9V to get more reasonable dimensions

Using assumptions from the previous part:

$$ro = 1.6M\Omega \quad , \quad V_{DS} = 0.6V \quad , \quad I_D = 6.25\mu A$$

Plugging in these Values in ADT SA we get:

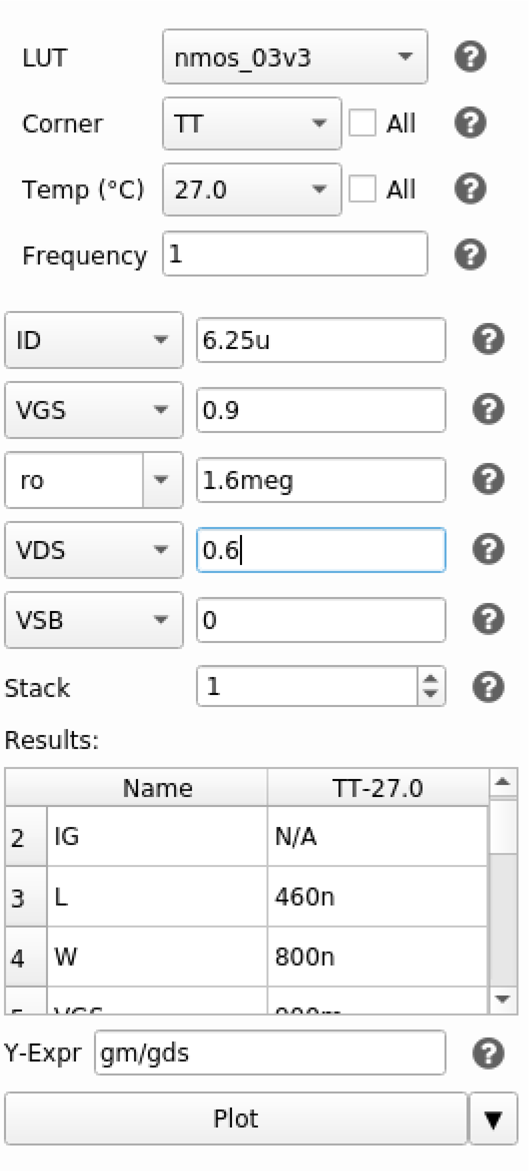


Figure 11 Initial Values for the PMOS Current Mirror Load Pair from SA

$$\boldsymbol{L = 460nm \quad , \quad W = 800nm}$$

Width might be a bit small so we can approximate it to about $\mathbf{1\mu}$ to account for CMIR-LOW later.

$$\boldsymbol{L = 460nm \quad , \quad W = 1\mu m}$$

Tail Current Mirror Sizing:

For the purposes of this design, I will use a simple current mirror at the tail.

Using the spec for CMRR:

$$CMRR = 74dB = A_{diff} - A_{CM} \rightarrow A_{CM} = -38dB$$

$$\rightarrow A_{CM} = 0.0126 \approx \frac{1}{2gm_{3,4}ro_5} \rightarrow \frac{1}{ro_5} = gds_5 \approx \mathbf{1.33\mu s}\ (Using\ Values\ of\ gm_{3,4}\ from\ SA)$$

Using the Spec for CMIR-Low:

$$CMIR_{HIGH} = 0.8V = V_{DD} - V_{GS1,2} - V_5^* \rightarrow V_5^* = 119.3mV \approx 120mV$$

$$Giving\ it\ a\ higher\ margin \rightarrow \boldsymbol{V_5^* \approx 150mV}$$

$$\boldsymbol{I_D = I_{B1} = 12.5\mu s} \rightarrow From\ previous\ assumption\ \text{of the input stage}$$

Plugging in these Values into ADT SA:

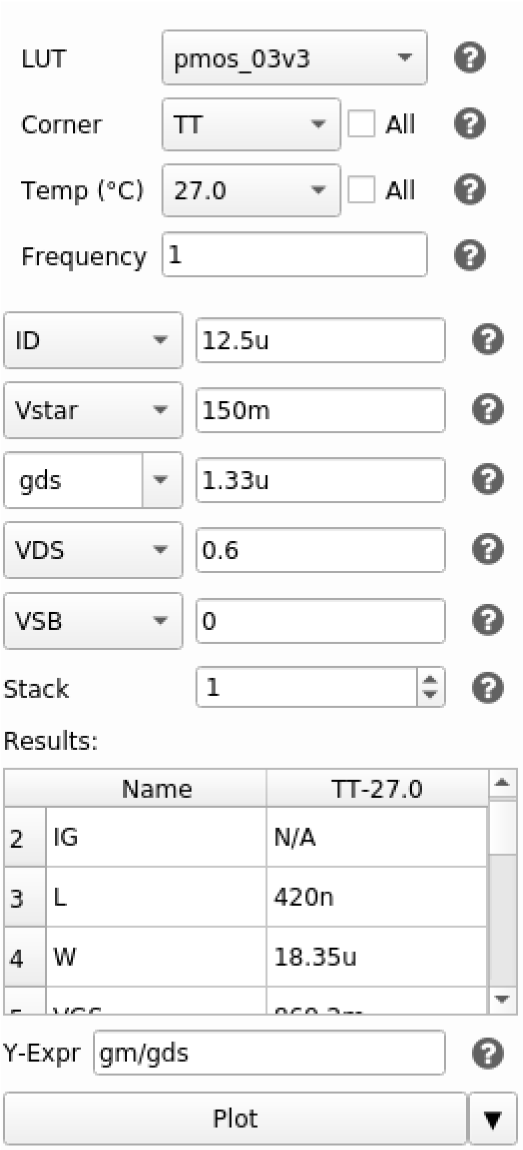


Figure 12 Initial Values for the Tail Current Mirror Load Pair from SA

$$\boldsymbol{L = 420nm} \quad , \quad \boldsymbol{W = 18.35\mu m}$$

Current Mirror:

The current tail mirror is easy to size, Using the sizing for the first mirror we can size the left and right one:

$$I_{B2} = 60\mu - 12.5\mu = 47.5 \approx 4I_{B1}$$

Thus, we can give the second current mirror the same sizing as M5 but with 4 multipliers. And sizing M8 (Left Transistor) is simple as well:

$$\frac{18.35\mu}{W} = \frac{12.5\mu}{10\mu} \rightarrow \boldsymbol{W = 14.68\mu \approx 14.5\mu}$$

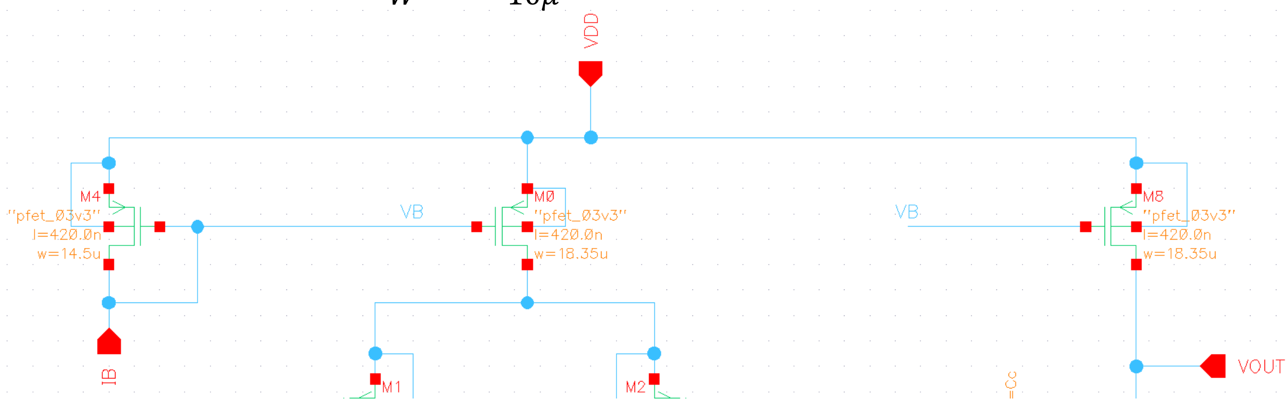


Figure 13 Tail Current Mirror

Second Stage CS Amplifier:

$$\boldsymbol{I_D = I_{B2} = 47.5\mu A}$$

And Putting $V_{DS} = \frac{V_{DD}}{2} = \mathbf{0.9V}$

$$A_v = 30dB \rightarrow 31.622 \approx \frac{gm_6 ro}{2}$$

Since the current mirror of the second stage uses the same sizing as the first stage one, we can its dimensions into ADT to get its value of gds

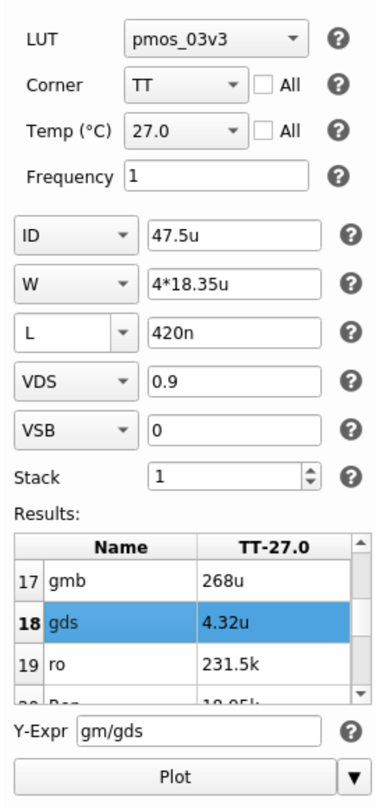


Figure 14 gds of Second Current Mirror

Moving on the value of gm for the second stage transistor can be calculated

$$A_v = 30dB \rightarrow 31.622 \approx \frac{gm_6 ro}{2} \rightarrow \boldsymbol{gm_6 = 273.2\mu S}$$

Plugging in every thing into ADT to get the dimensions:

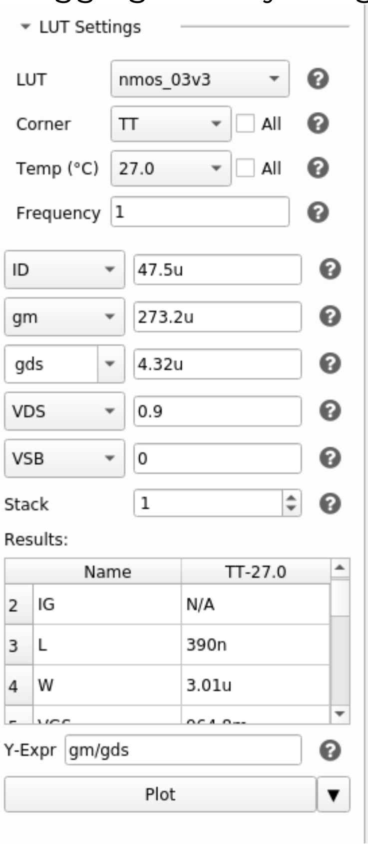


Figure 15 Sizing for CS Second Stage Transistor

$$\boldsymbol{L = 390nm \quad , \quad W = 3.01\mu m}$$

Using this though would result in the circuit latching due to the output swing of the first stage not matching the input of the second stage (Systematic Mismatch) thus the node between the two stages is floating and the voltage is not set, hence the second current mirror gets out of saturation

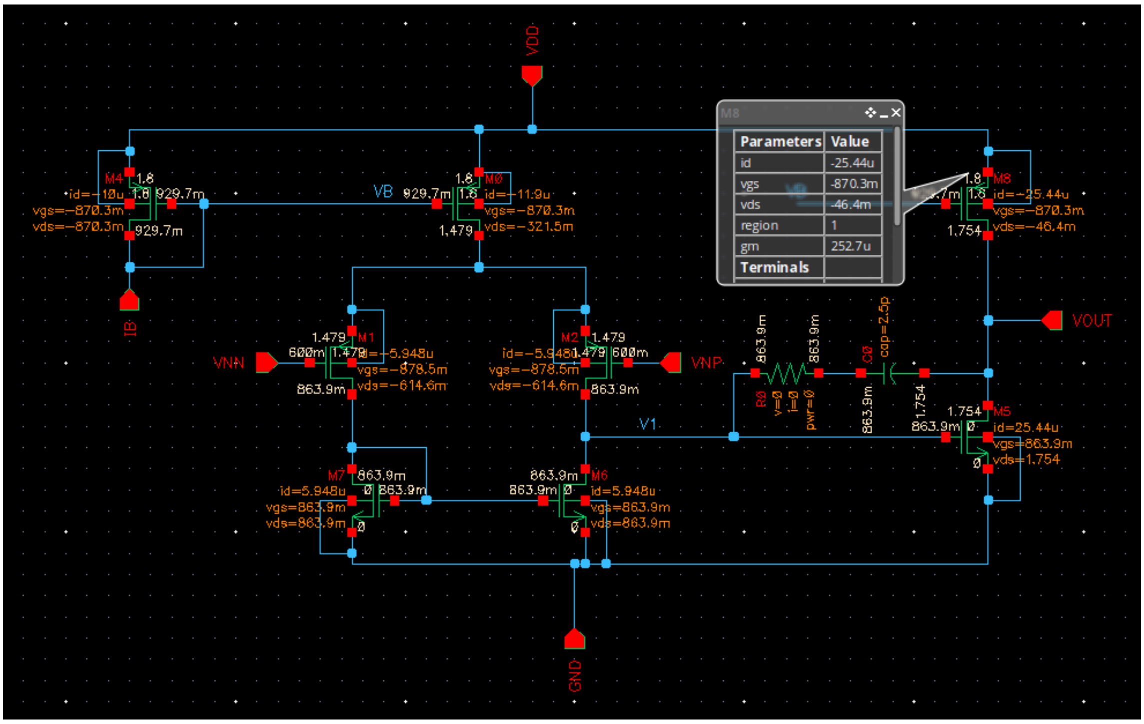


Figure 16 Circuit Latching using the calculated dimensions

To fix this we can either resize the NMOS Current Mirror Load with the new value of VGS at VDS or just sweep W for the CS transistor till the current source gets into saturation.
And to take the Output Swing Spec into consideration we will also look for the point exactly where VOUT=0.9V for the best result.

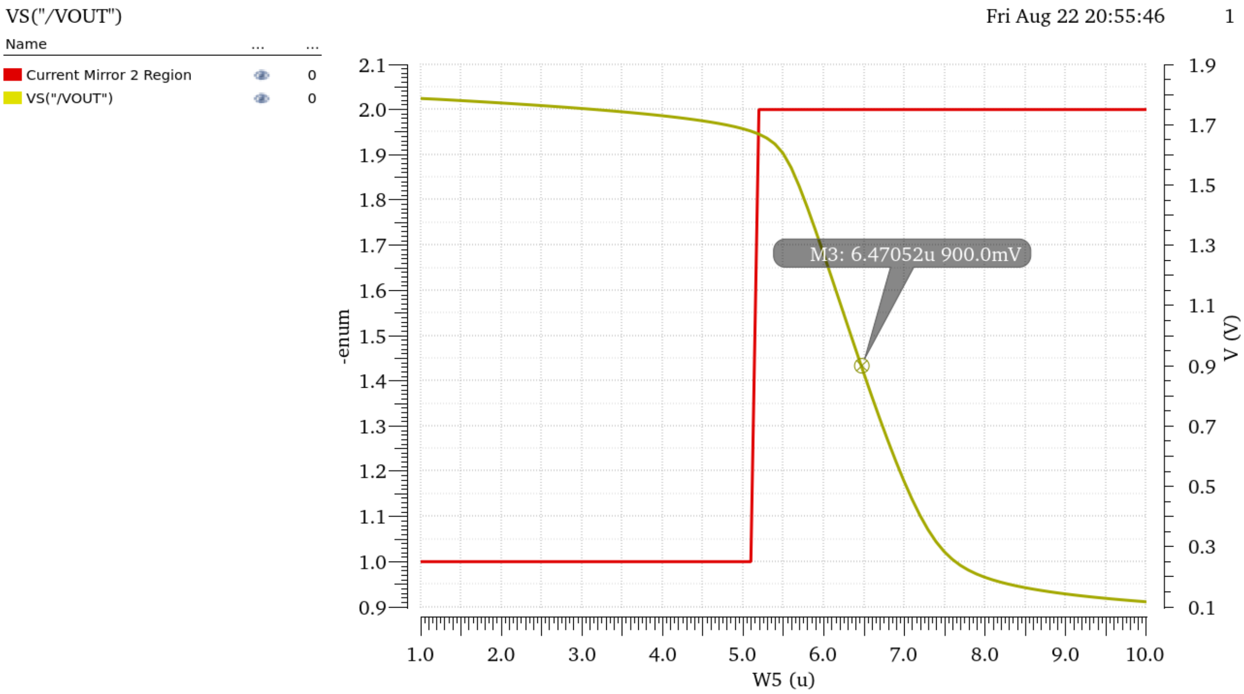


Figure 17 Point at which Vout=0.9V and Current mirror in Sat

$$L = 390nm \quad , \quad W = 6.47\mu m$$

Feedforward Zero:

Finally, finding the value of the resistor to cancel the feedforward zero is simple enough:

$$R_C = \frac{1}{gm_6} \approx 2.5K\Omega$$

Initial Design Point:

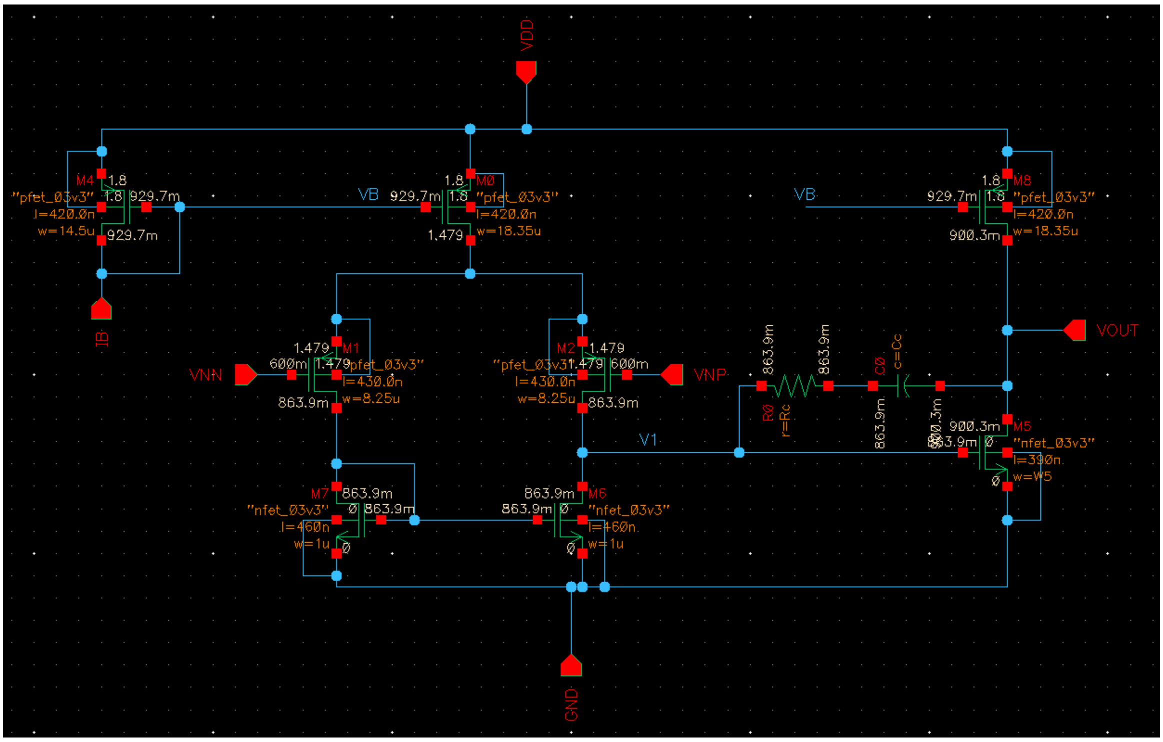


Figure 18 Final Design Schematic

	M1,2	M3,4	M5	M6	M7	M8
W	8.25um	1um	18.35um	6.47um	18.35	14.5
Multiplier	1	1	1	1	4	1
gm	8E-05	5E-05	0.0003	0.0004	0.0006	0.0001332
L	430nm	460nm	420nm	390nm	420nm	420nm
ID	6.25uA	6.25uA	12.5uA	47.5uA	47.5uA	10uA
gm/ID	12.65	8.658	20.17	8.552	13.46	13.32
VDsat	134.4mV	193.6mV	68.72mV	189.5mV	124.5mV	125.6mV
Vov	100.1mV	168.4mV	46.76mV	165.8mV	84.94mV	86.54mV
Vstar	158.1mV	231mV	99.16mV	233.9mV	148.6mV	150.1mV

M1,2: Input Transistors (PMOS)
M3,4: NMOS Current Mirror Load (NMOS)
M5: Tail Current Source of First Stage (PMOS)
M6: Input Transistor of Second Stage (NMOS)
M7: Tail Current Source of Second Stage (PMOS)
M8: Current Mirror (PMOS)

Simulation and Design Iteration:

Of Course, this design wouldn't satisfy all the specs exactly from the first run, thus I'll change them as we go through the requirements and tweak slightly till all specs are met

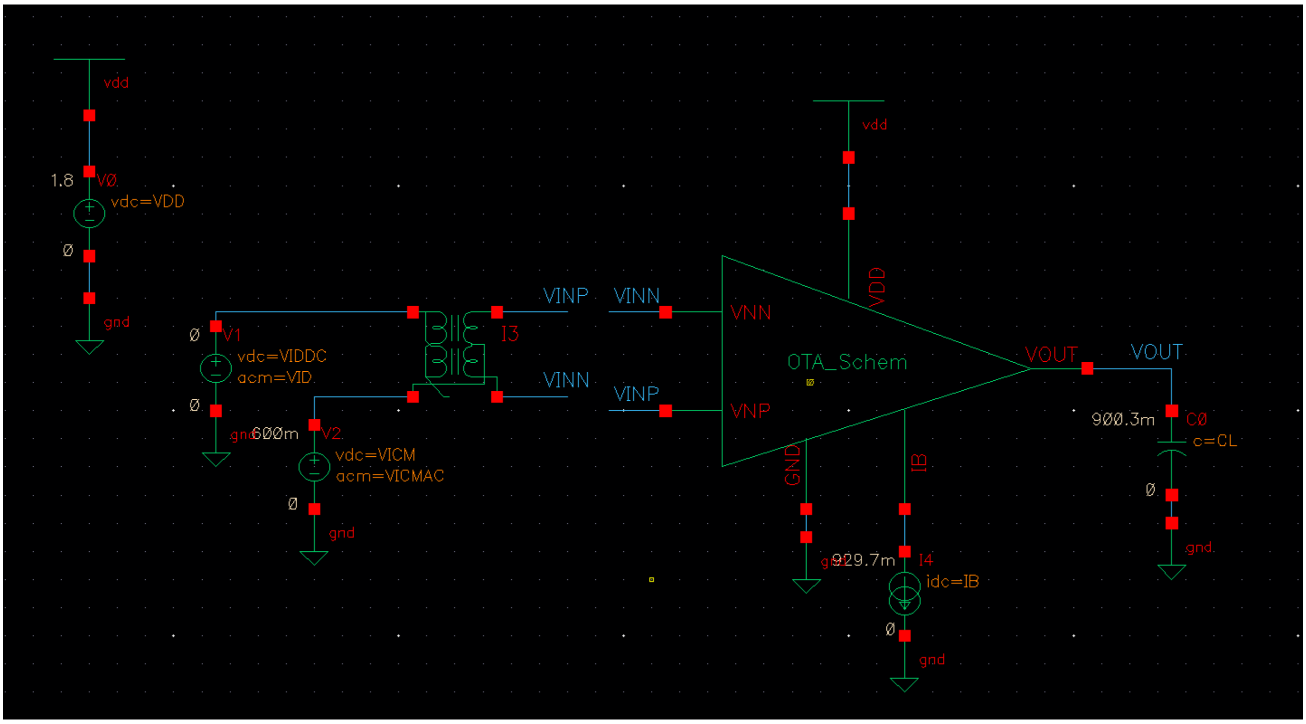


Figure 19 Testbench Schematic

Part 3: Open Loop Simulation:

OP Analysis:

```
BSIM4v5: Berkeley Short Channel IGFET Model-4
    device           m.x1.xm6.m0           m.x1.xm7.m0           m.x1.xm8.m0
     model            nmos_3p3.8           pmos_3p3.12           pmos_3p3.12
        id           5.07136e-05           5.07136e-05                 1e-05
        gm           0.000423818           0.000680299             0.0001342
      gmbs           0.000132045           0.000286348           5.64929e-05
       gds           5.64943e-06           4.42576e-06           8.88304e-07
       vgs              0.863887              0.870294              0.870294
       vth              0.691847              0.782372              0.782447
       vds              0.896256              0.903723              0.870294
     vdsat              0.193733              0.125873              0.125822

BSIM4v5: Berkeley Short Channel IGFET Model-4
    device           m.x1.xm5.m0           m.x1.xm4.m0           m.x1.xm3.m0
     model           pmos_3p3.12            nmos_3p3.4            nmos_3p3.4
        id           1.18956e-05           5.94782e-06           5.94782e-06
        gm           0.000160388           5.19909e-05           5.19909e-05
      gmbs           6.76244e-05           1.80182e-05           1.80182e-05
       gds            1.9202e-06           5.37556e-07           5.37556e-07
       vgs              0.870295              0.863889              0.863889
       vth               0.78369               0.69068               0.69068
       vds              0.321492              0.863882              0.863882
     vdsat              0.124984               0.19818               0.19818
```

```
BSIM4v5: Berkeley Short Channel IGFET Model-4
    device           m.x1.xm1.m0           m.x1.xm2.m0
     model            pmos_3p3.8            pmos_3p3.8
        id           5.94782e-06           5.94782e-06
        gm            7.7504e-05            7.7504e-05
      gmbs           3.29316e-05           3.29316e-05
       gds           5.92137e-07           5.92137e-07
       vgs              0.878506              0.878506
       vth              0.783656              0.783656
       vds              0.614608              0.614608
     vdsat              0.130231              0.130231
```

Figure 20 OP Point from Xschem

I used xschem and ngspice to get the operating point as it displays in a much better way than cadence, the values are largely almost identical though.

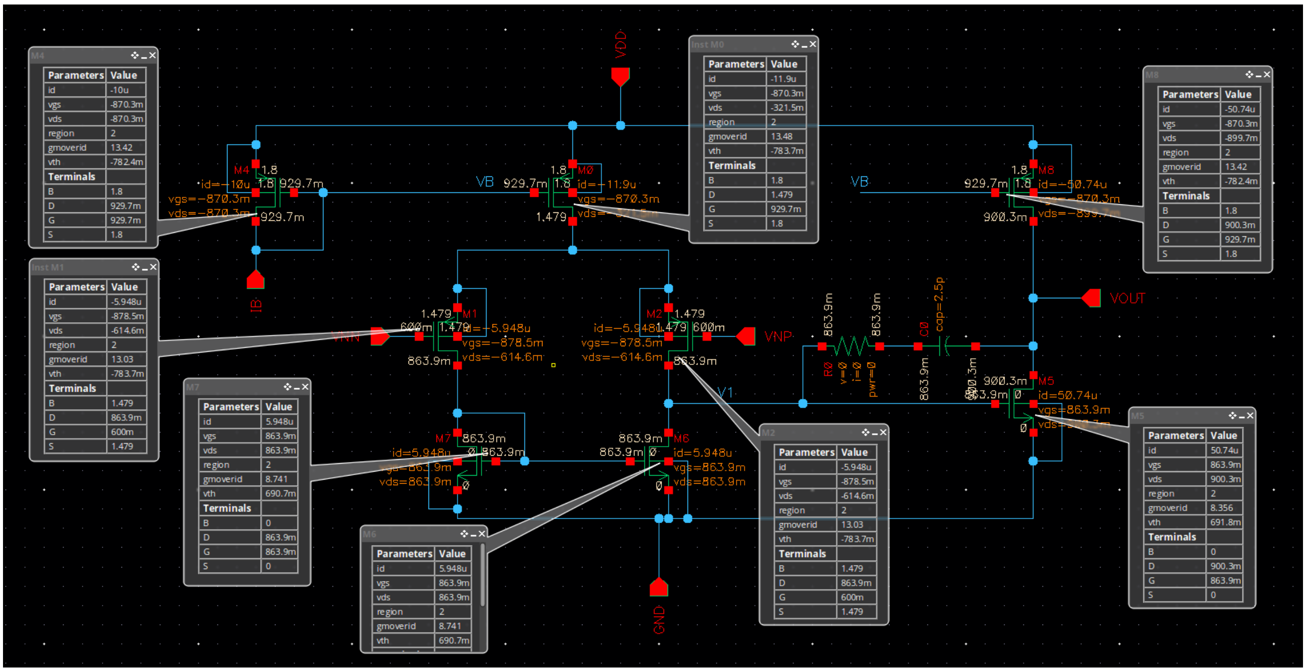


Figure 21 OTA Schematic with OP Point and DC Voltages Annotated

• Is the current (and gm) in the input pair exactly equal?

Yes, the current is exactly equal in both transistors of the input pair.

• What is DC voltage of the first stage? Why?

$$V_{out} = 863.9\text{mV}$$
$$V_{out} = V_{DD} - V_{DS1,2} - \mathrm{V}_{\mathrm{DS5}} = 1.8 - 0.6146 - 0.321 = \mathbf{0.864}\boldsymbol{V}$$

• What is DC voltage of the second stage? Why?

$$V_{out} = 0.9\text{V}$$
$$V_{out} = V_{DD} - V_{DS7} = 1.8 - 0.9 = \mathbf{0.9}\boldsymbol{V}$$

Systematic mismatch is pretty low as seen here between the vout of both stages.

Differential Small Signal:

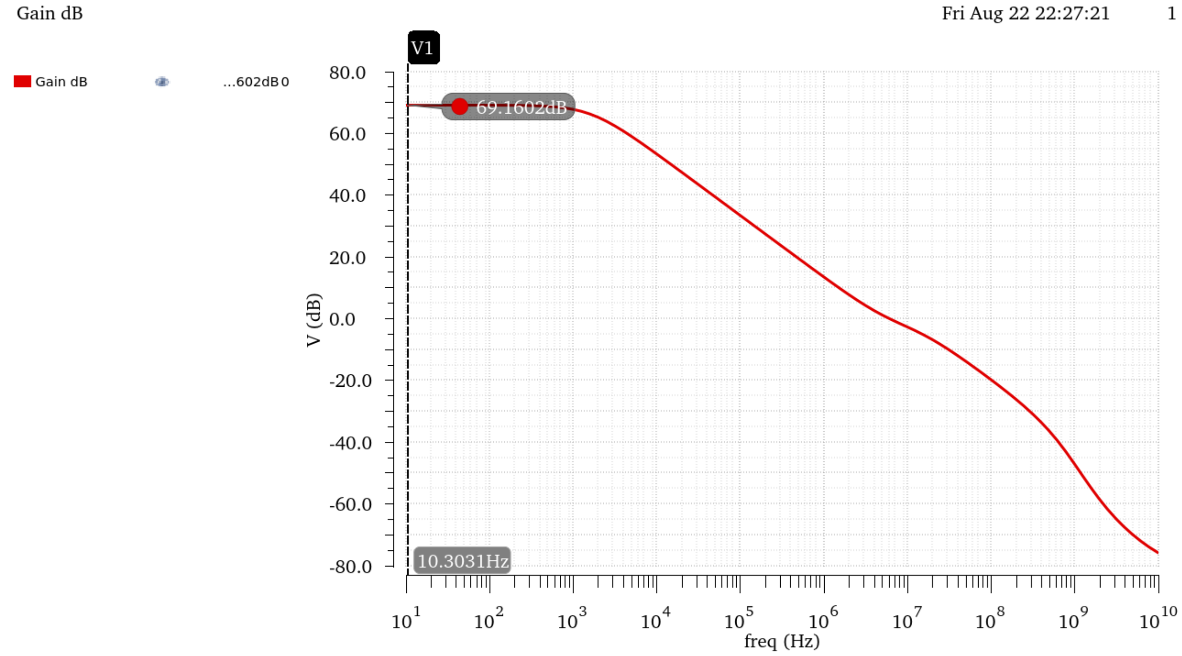


Figure 22 Diff Gain (dB) Open Loop Gain vs Frequency

BW	1.646k
fu	6.202M
GBW	4.737M
Ao dB	69.16
Ao	2.871k

Figure 23 Diff Small Signal Outputs

Hand Analysis:
Using Values from OP Analysis:

$$First\ Stage\ Gain: |\boldsymbol{A_{vd}}| = \boldsymbol{gm_{1,2}(ro_{1,2}//ro_{3,4}) = 68.644 \rightarrow 36.732\ dB}$$

$$Second\ Stage\ Gain: \boldsymbol{A_v = gm_6(ro_6//ro_7) = 42.06 \rightarrow 32.5dB}$$

$$Overall\ Gain: \boldsymbol{A_v = 36.732 + 32.5 = 69.21 \rightarrow 2887.31}$$

$$\boldsymbol{GBW = \frac{gm_1}{2\pi C_c} = 4.9338MHz}$$

	Simulation	Hand Analysis
Gain	2871	2887
Gain (dB)	69.16	69.21
GBW	4.737MHz	4.9338MHz

AOL Gain exceeds the required spec! and agrees with hand analysis.
GBW In hand analysis is a bit less than simulation and less than the required spec as our analysis did not account for other circuit and MOS capacitances, this can be fixed by simply lowering Cc a little bit.

BW	1.79k
fu	6.608M
GBW	5.152M
Ao dB	69.16
Ao	2.871k

Figure 24 Simulation Results after Lowering Cc to 2.3pF

Lowering Cc results in giving us the required GBW from the spec, but it might affect our phase margin, this will be fine-tuned further in Part 4.

CM Small Signal:

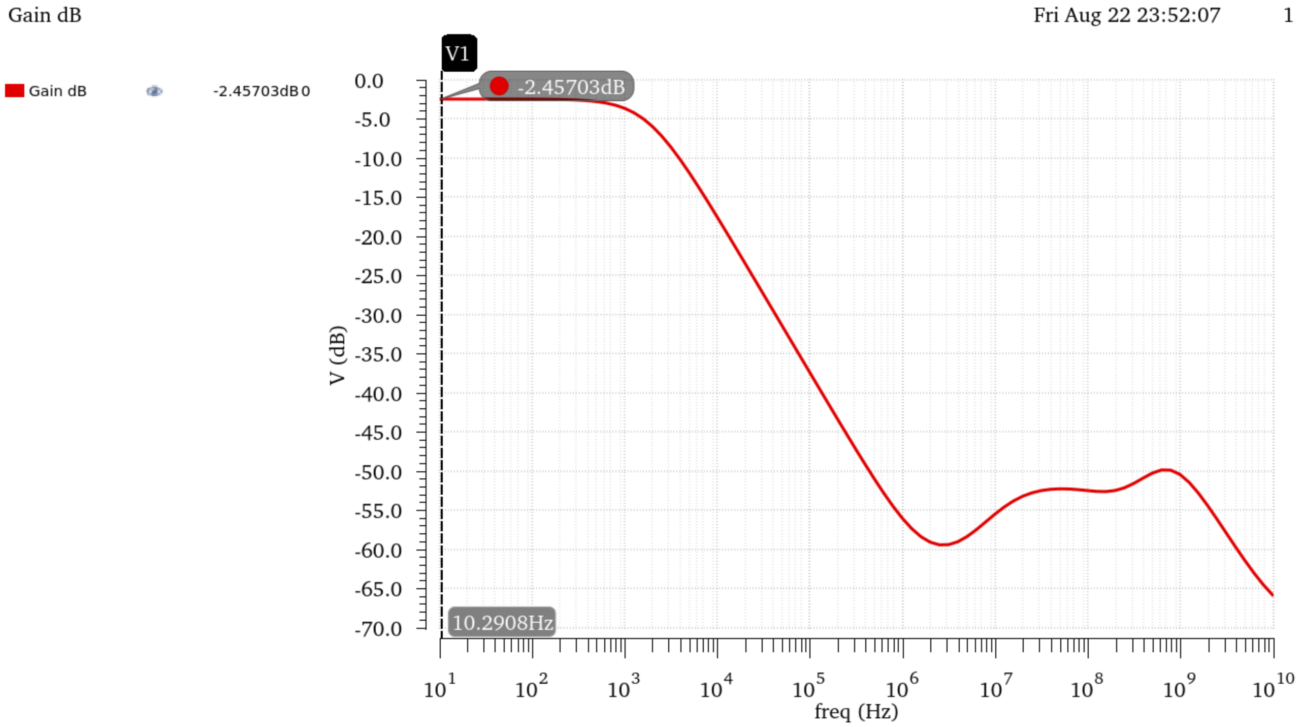


Figure 25 CM Gain (dB) vs Frequency

BW	1.79k
GBW	1.352k
Ao dB	-2.457
Ao	753.6m

Figure 26 CM Gain Values

Hand Analysis:
Using Values from OP Analysis:

$$|A_{vCM}| = \frac{1}{2gm_{3,4}ro_5} * gm_6(ro_6//ro_7) = 776.65 \times 10^{-3} \rightarrow -2.195\ dB$$

	Simulation	Hand Analysis
Gain	753.6e-3	775.65e-06
Gain (dB)	-2.457	-2.195

Hand analysis and simulation results agree with each other.

CMRR (XF Simulation):

$$CMRR = A_d - A_{cm} \approx 71.405dB$$

CMRR is slightly less than the spec, this can be fixed by slightly increasing the CM Gain which could be achieved by increasing the length of the tail current sources slightly.

I'll fine tune this using XF simulation.

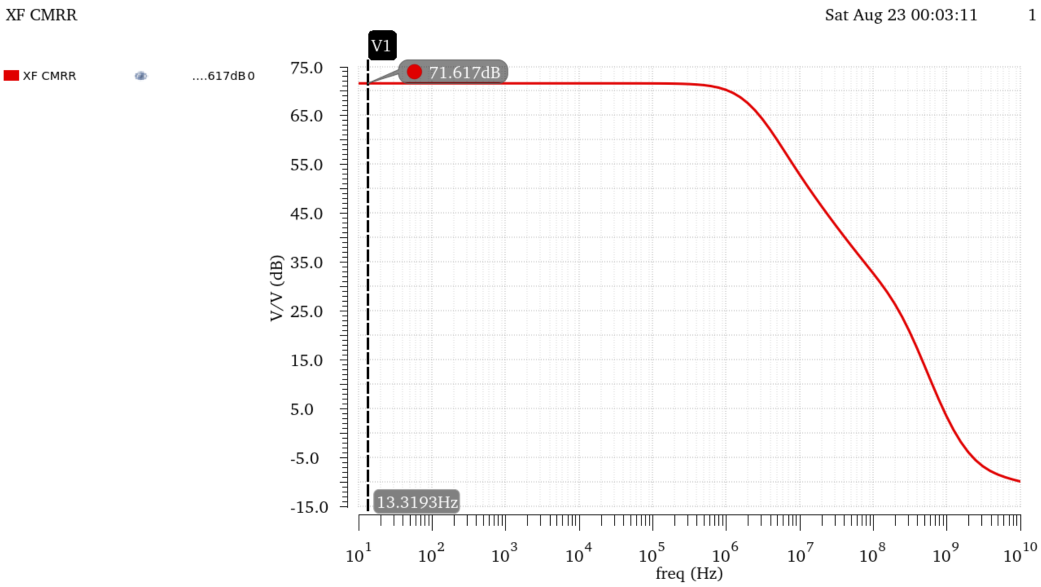


Figure 27 CMRR From XF Simulation

CMRR	71.62

Figure 28 CMRR Value from Simulation

	Simulation	Hand Analysis
CMRR (dB)	71.62	71.405

As expected, CMRR was a little less than the required spec, I'll fixed this by slightly increasing the Lengths of the tail current sources, increasing their seen resistance and increasing CM Gain.

After Fine Tuning:

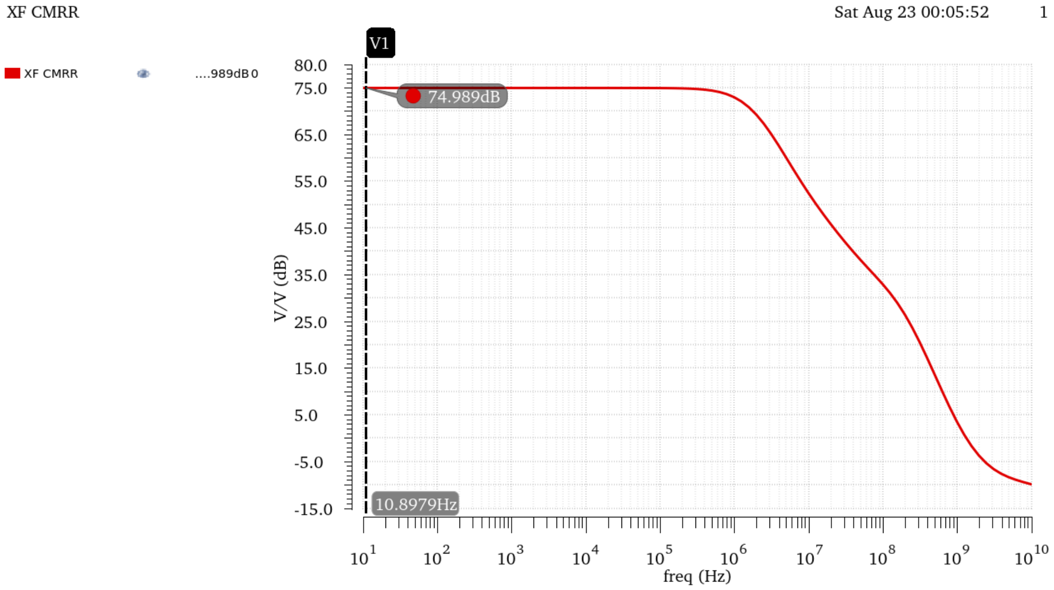


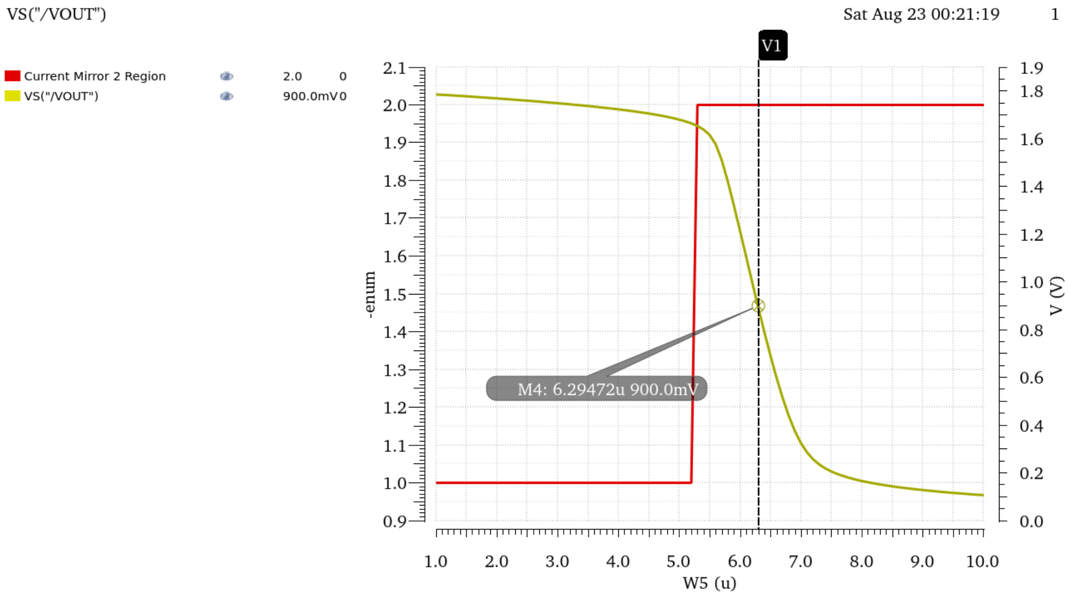
Figure 29 CMRR After Fine Tuning

CMRR	74.99

Figure 30 CMRR Value after Tuning

$$using\ \boldsymbol{L_{5,7,8}} = \boldsymbol{600nm}\ instead\ of\ 420nm$$

This changed the current slightly in the CS Stage, thus we'll have to resize W6 to ensure Vout is @ 0.9V



$$\boldsymbol{W_{6new} = 2.295\mu m}$$

All other design points stay mostly the same!

Diff Large Signal:

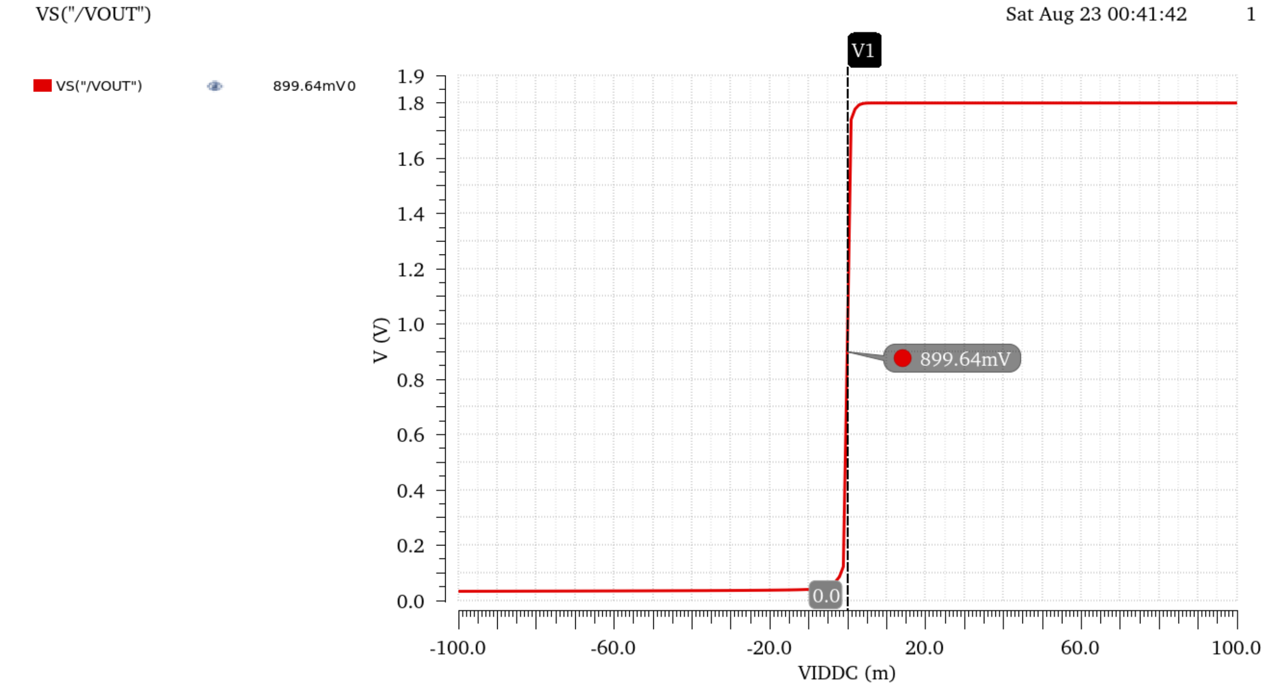


Figure 31 VOUT vs Vid

What is the value of Vout at VID = 0? Why?

$\boldsymbol{Vout_{@Vid=0} = 0.9V,}$ Same Value as Vout calculated During OP Analysis, This is the Quiescent point of the amplifier on which we biased our transistor for the maximum output swing

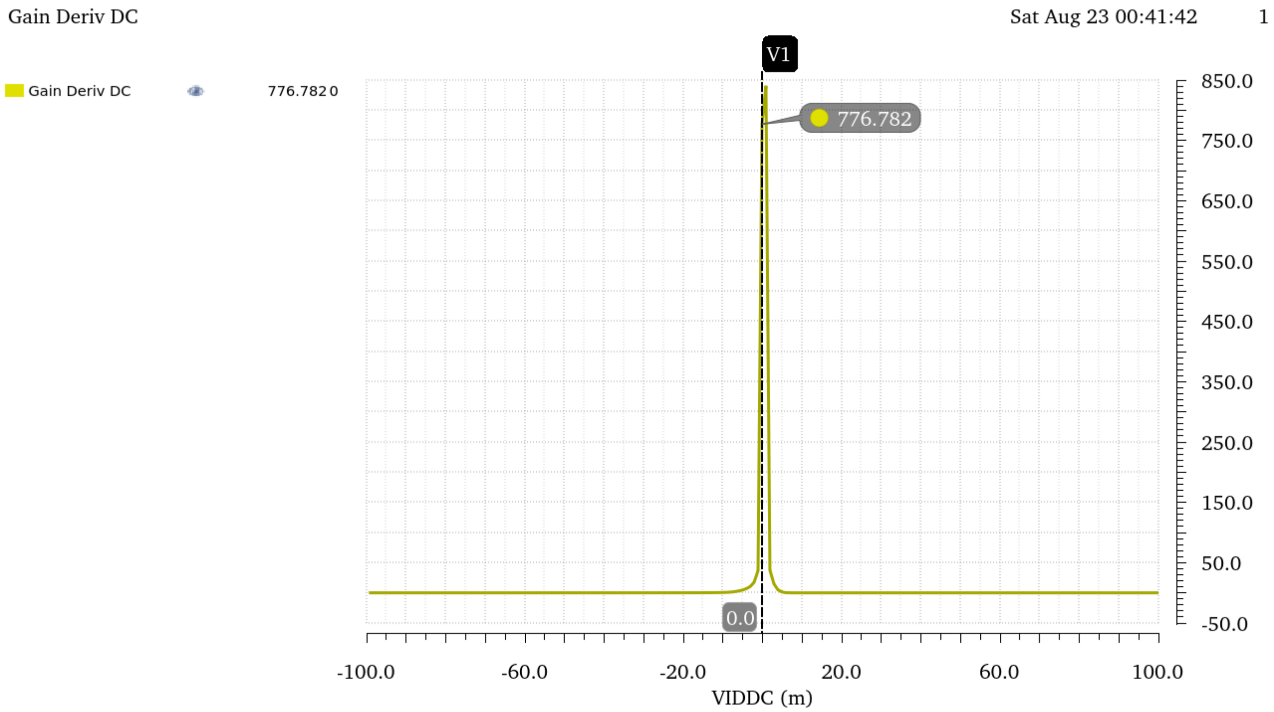


Figure 32 Derivates of Vout vs Vid

Peak of Derivative	840.5

Figure 33 Value at the Peak

- **Is the peak less than the value of Avd obtained from ac analysis? Why?**

The Value of the derivative of Vout vs Vid is the DC Gain, the value at the peak=**840.5** is observed to be way lower than Avd=**2871**

This is due to the very high gain of the amplifier such that the linear region in which the derivative is meaningful is very small, the 1mV step required in the question is just not enough to measure the fine details and values to get the actual gain, This can be fixed by using a lower step size but at the cost of higher processing power and longer simulation time.

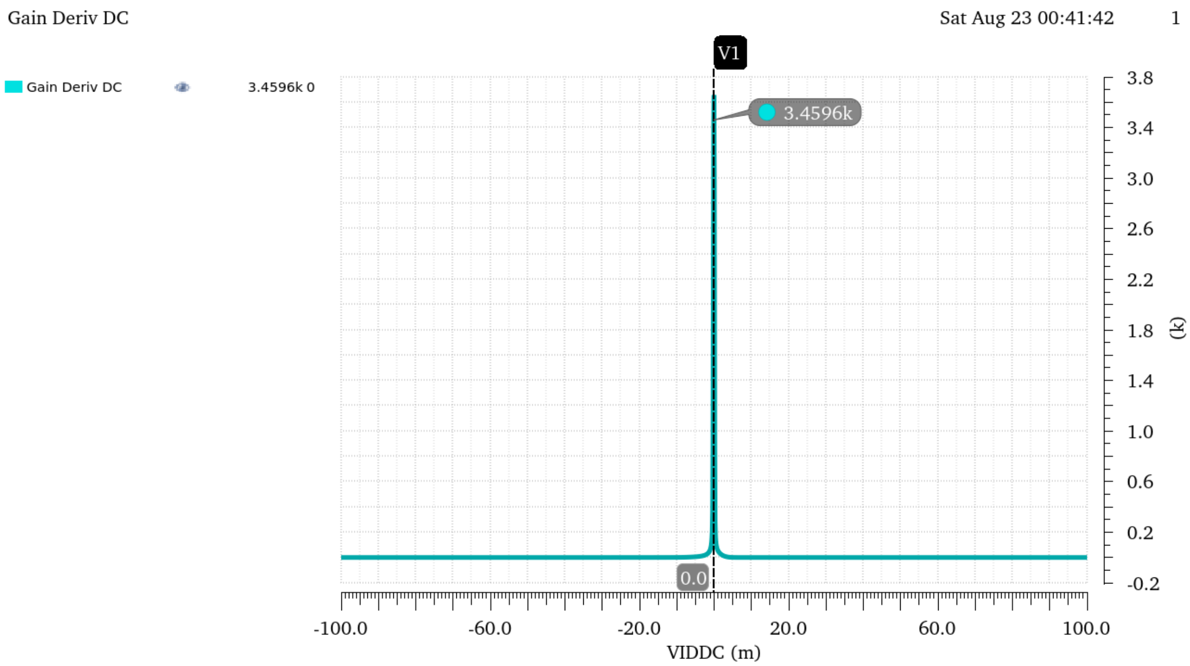
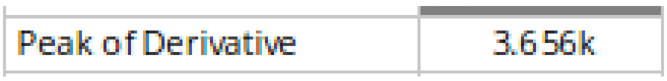


Figure 34 Derivative of Vout vs VID with a step of 0.1mV instead of 1mV



Peak of Derivative	3.656k

Figure 35 New Value at the Peak after using finer step

Using a finer step resulted in a much better comparable value to the Gain obtained from AC Analysis!
Notice the gain is also slightly higher than what was calculated earlier, this is due to the increase of RSS after changing the lengths of the current sources to achieve the required CMRR.

CM Large Signal:

Region vs VICM:

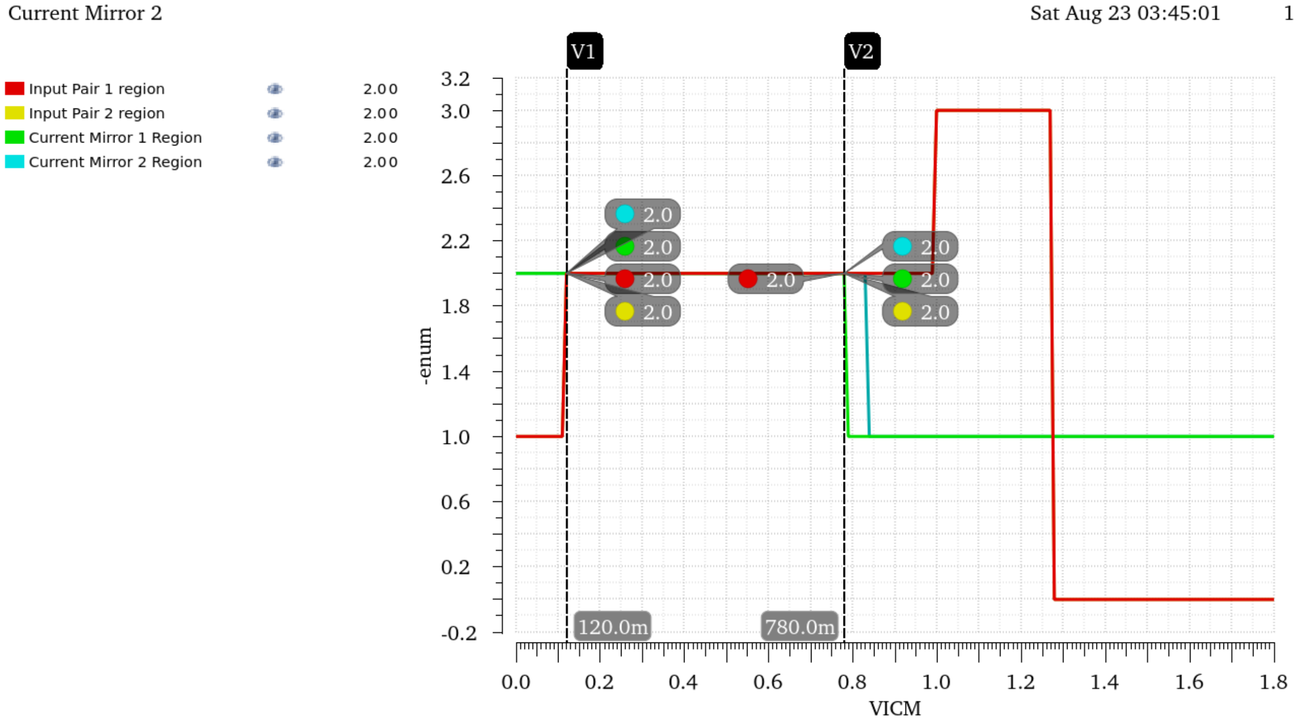


Figure 36 Region vs VICM

$$VICM_{LOW} = 120mV\ ,\quad VICM_{HIGH} = 780mV$$

We can observe that the Lower CMIR is met but the Higher Range is close but not met exactly,

Fine Tuning:

This can be remedied by increasing the Widths of the tails current sources slightly or increasing the Widths of the input stages, I will also decrease the widths of the Current Mirror transistors to reduce current as I'm currently a little bit over the power consumption spec.

While I'm fine tuning those, I'll keep an eye on VOUT to make the necessary changes for W6 to keep it at 0.9V as well as the other design parameters to ensure they are within spec.

Final Values after fine Tuning:

$$W_{1,2} = 10.3\mu m \quad , \quad W_{5,7} = 17.3\mu m \quad , W_6 = 6.26\mu m$$

GBW	5.444M
Diff Gain	71.97
CM Gain	-4.566
CMRR	76.54

Figure 37 Design Specs after change

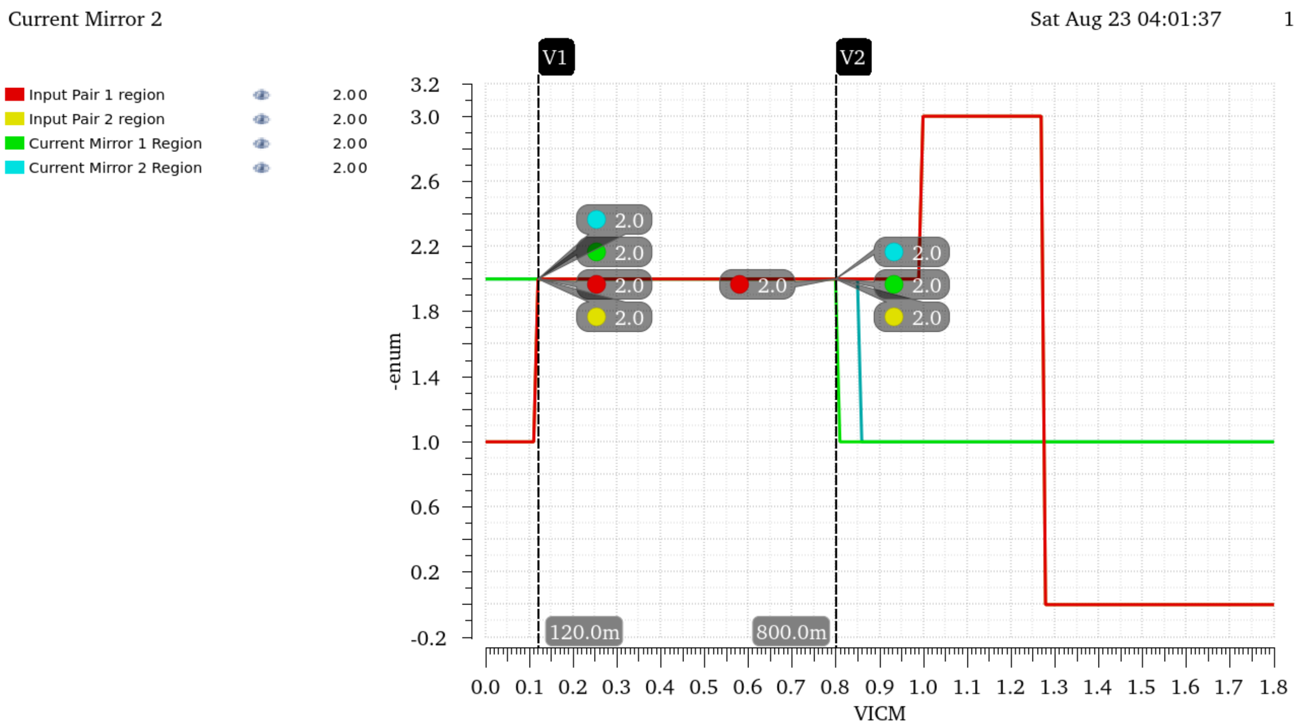


Figure 38 Regions vs VICM After Fine Tuning

$$VICM_{LOW} = 120mV \, , \quad VICM_{HIGH} = 800mV$$

As we can see the slight increase in width improved other parameters as well.

(GBW vs VICM):

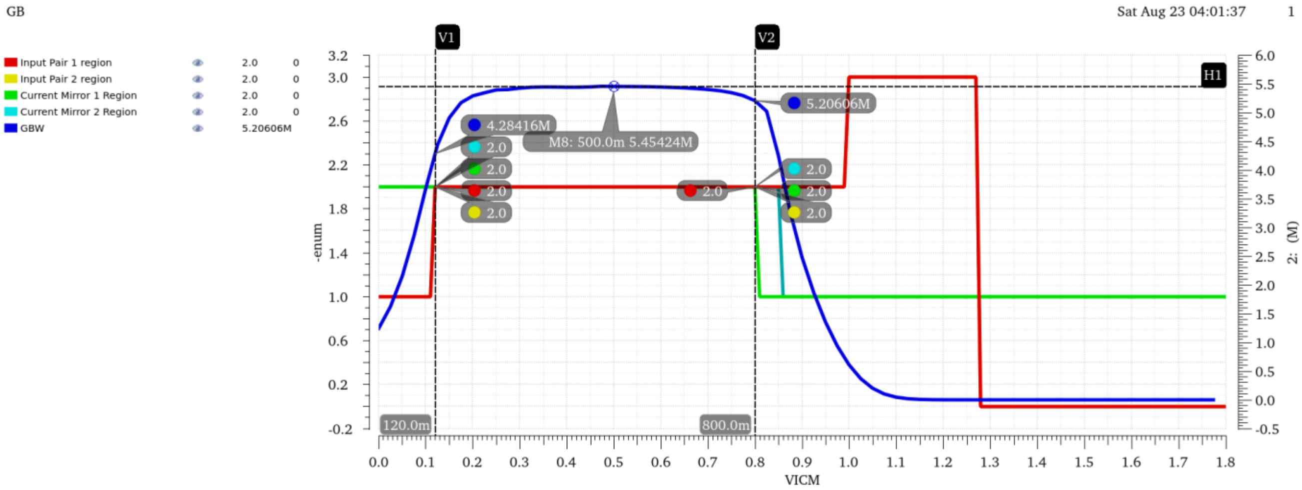


Figure 39 GBW and Regions Annotated and Overlaid

The previous figure shows the CMIR as illustrated by the regions

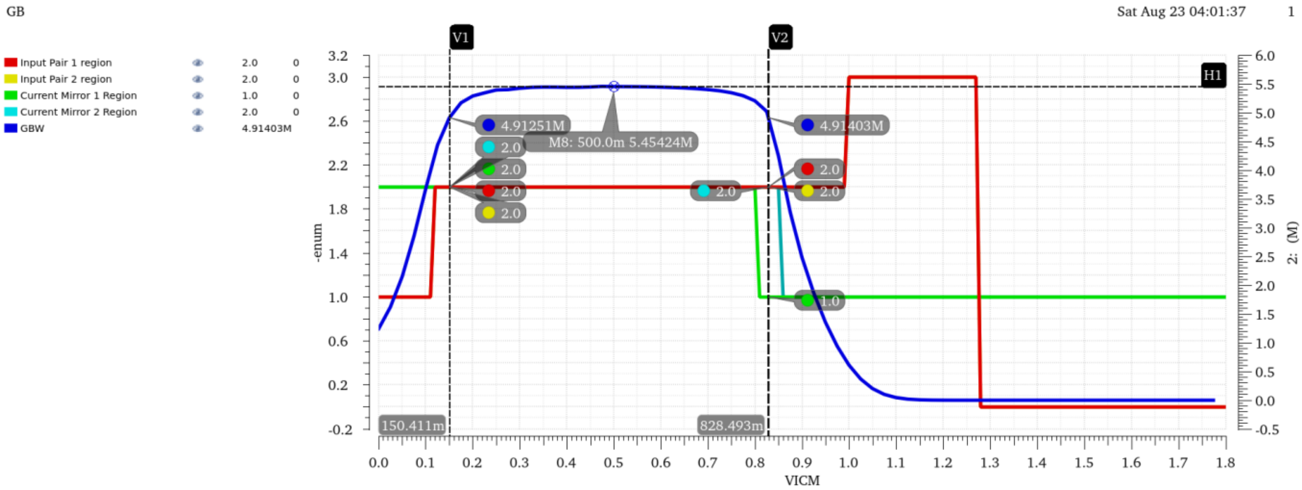


Figure 40 GBW and Region Overlaid and Annotated @ GBW=90% Max

The previous one showcases the CMIR at range where GBW is within 90% if its max value (Also Annotated).

$$\boldsymbol{VICM_{LOW} = 150.411mV\ ,\quad VICM_{HIGH} = 828.49mV}$$

The design exceeds the required specs.

Hand Analysis:

Calculating the CMIR analytically is quite tricky as some of the device operate in regions where the square law isn't valid, but we can work around it to get rough estimates.

$$VICM_{LOW} = VGS_{3,4} - V_{TH1,2} \approx Vov_{3,4} \approx \mathbf{160mV}$$
$$VICM_{HIGH} = V_{DD} - V_{GS1,2} - Vov_5 \approx \mathbf{816.6mV}$$

	Simulation	**Hand Analysis**
CMIR-LOW	150.411mV	160mV
CMIR-HIGH	828.48mV	816.6mV

The results agree with the hand analysis for the most part.

Final Sizing at the end of Part 3:

Since I've fine-tuned the design a lot throughout the procedure, here's a summary of the new sizing for each transistor:

	M1,2	M3,4	M5	M6	M7	M8
W	10.3um	1um	17.3um	6.26um	17.3um	14.5
Multiplier	1	1	1	1	4	1
L	430nm	460nm	600nm	390nm	600nm	600nm
gm	8.14E-05	5.11E-05	1.32E-04	4.04E-04	5.48E-04	1.15E-04
ID	5.77E-06	5.77E-06	1.15E-05	4.76E-05	4.76E-05	1.00E-05
gm/Id	1.41E+01	8.86E+00	1.15E+01	8.48E+00	1.15E+01	1.15E+01
Vstar	1.42E-01	2.26E-01	1.74E-01	2.36E-01	1.74E-01	1.74E-01
Vov	7.61E-02	1.70E-01	1.25E-01	1.69E-01	1.25E-01	1.25E-01
Vdsat	1.18E-01	1.96E-01	1.43E-01	1.91E-01	1.44E-01	1.44E-01

```
BSIM4v5: Berkeley Short Channel IGFET Model-4
    device          m.x1.xm6.m0          m.x1.xm7.m0          m.x1.xm8.m0
     model           nmos_3p3.8         pmos_3p3.13         pmos_3p3.13
        id          4.76467e-05          4.76467e-05              1e-05
        gm          0.000403753          0.000547691          0.000114944
      gmbs          0.000125791          0.000250268          5.25237e-05
       gds          5.35827e-06          2.00264e-06          4.16313e-07
       vgs             0.860406             0.915428             0.915428
       vth             0.691782             0.790191             0.790187
       vds             0.897256             0.902723             0.915427
     vdsat             0.191378             0.143565             0.143567

BSIM4v5: Berkeley Short Channel IGFET Model-4
    device          m.x1.xm5.m0          m.x1.xm4.m0          m.x1.xm3.m0
     model          pmos_3p3.13           nmos_3p3.4           nmos_3p3.4
        id          1.15329e-05          5.76644e-06          5.76644e-06
        gm          0.000132483          5.11556e-05          5.11556e-05
      gmbs          6.05468e-05          1.77265e-05          1.77265e-05
       gds          1.12952e-06          5.25798e-07          5.25798e-07
       vgs             0.915428             0.860408             0.860408
       vth             0.790355             0.690687             0.690687
       vds             0.339716             0.860401             0.860401
     vdsat              0.14345             0.195581             0.195581

BSIM4v5: Berkeley Short Channel IGFET Model-4
    device          m.x1.xm1.m0          m.x1.xm2.m0
     model          pmos_3p3.12          pmos_3p3.12
        id          5.76645e-06          5.76645e-06
        gm          8.14682e-05          8.14682e-05
      gmbs          3.46136e-05          3.46136e-05
       gds          5.94916e-07          5.94916e-07
       vgs             0.860282             0.860282
       vth              0.78415              0.78415
       vds             0.599866             0.599866
     vdsat             0.117637             0.117637
```

Figure 41 Updated OP Point

Part 4: Closed Loop OTA Simulation:

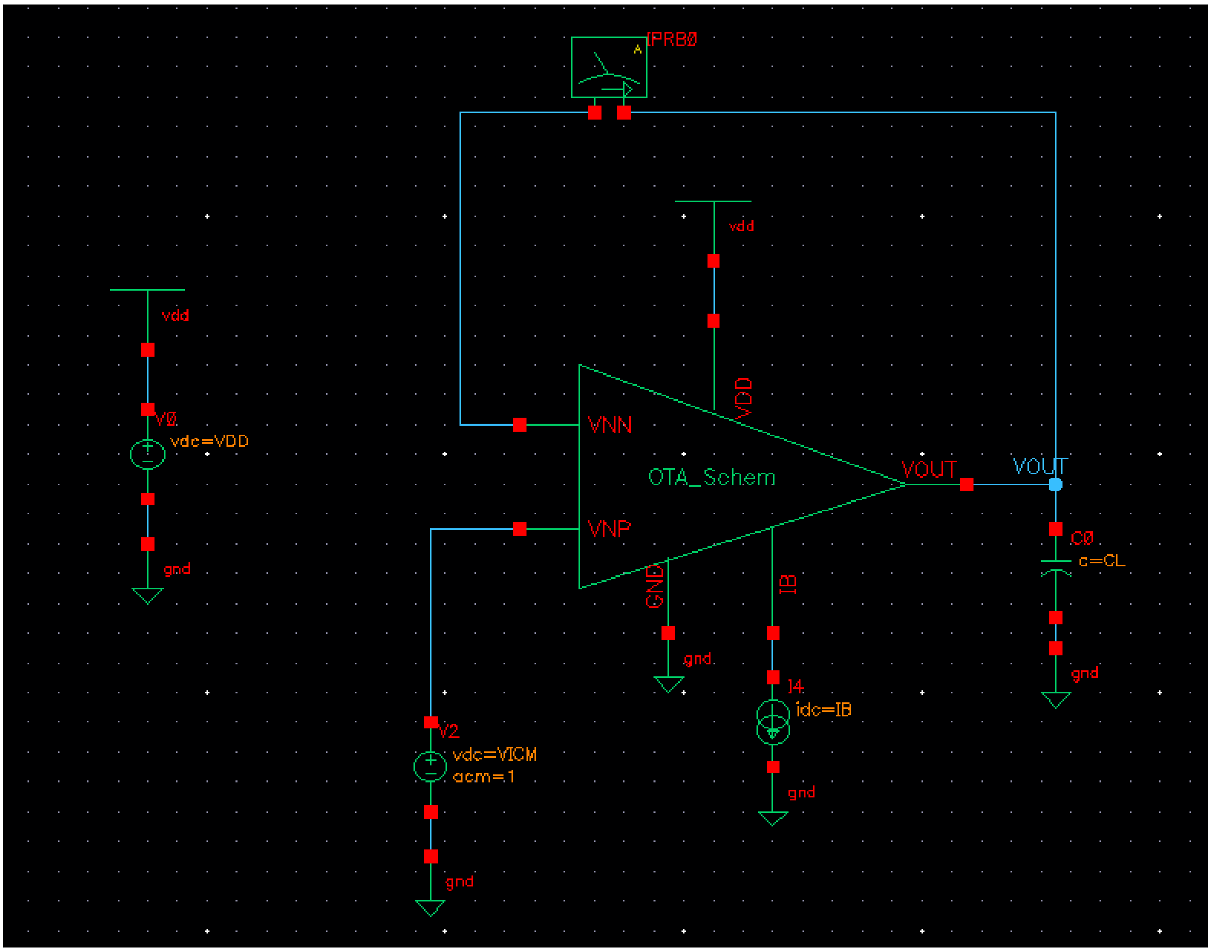


Figure 42 Closed Loop Testbench

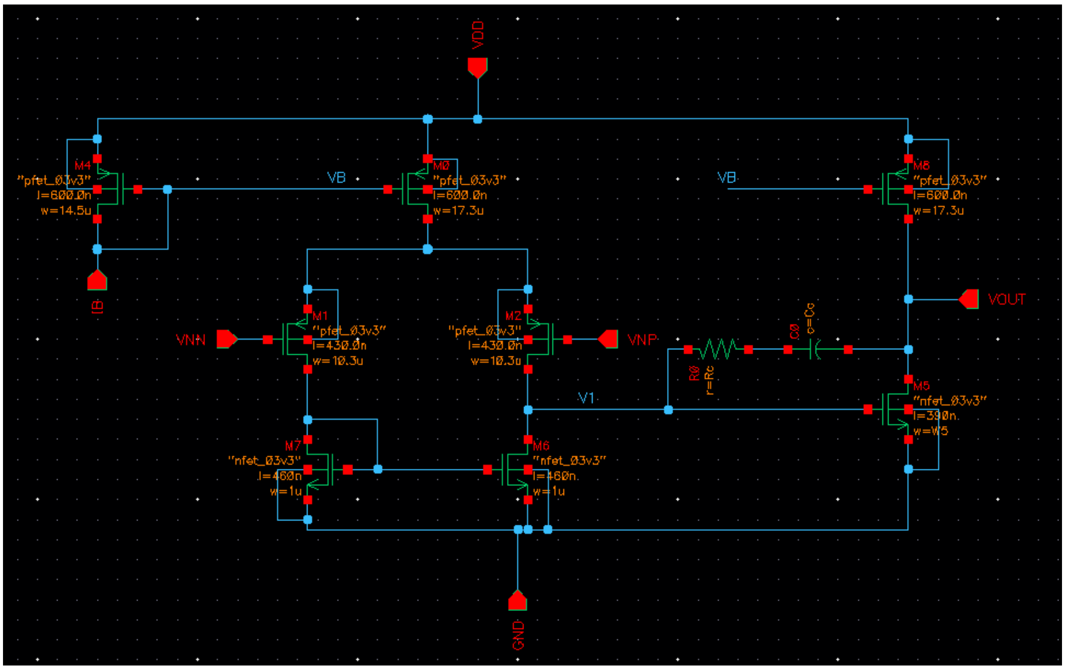


Figure 43 OTA Schematic

Op Analysis:

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm6.m0	m.x1.xm7.m0	m.x1.xm8.m0
model	nmos_3p3.8	pmos_3p3.13	pmos_3p3.13
id	4.81888e-05	4.81888e-05	1e-05
gm	0.000402988	0.000553426	0.000114944
gmbs	0.00012574	0.000252864	5.25237e-05
gds	6.62378e-06	1.67424e-06	4.16313e-07
vgs	0.86607	0.915428	0.915428
vth	0.693253	0.790104	0.790187
vds	0.600059	1.19992	0.915427
vdsat	0.194276	0.143625	0.143567

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm5.m0	m.x1.xm4.m0	m.x1.xm3.m0
model	pmos_3p3.13	nmos_3p3.4	nmos_3p3.4
id	1.15328e-05	5.76791e-06	5.76492e-06
gm	0.000132482	5.11694e-05	5.11485e-05
gmbs	6.05464e-05	1.77312e-05	1.7724e-05
gds	1.12978e-06	5.23879e-07	5.25699e-07
vgs	0.915428	0.860378	0.860378
vth	0.790356	0.690676	0.690687
vds	0.339657	0.866066	0.860371
vdsat	0.14345	0.195568	0.195559

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm1.m0	m.x1.xm2.m0
model	pmos_3p3.12	pmos_3p3.12
id	5.76492e-06	5.76791e-06
gm	8.14532e-05	8.14727e-05
gmbs	3.46072e-05	3.4616e-05
gds	5.94727e-07	5.97972e-07
vgs	0.860262	0.860341
vth	0.78415	0.784161
vds	0.599954	0.59426
vdsat	0.117625	0.117668

Figure 44 OP Point from Xschem and Ngspice

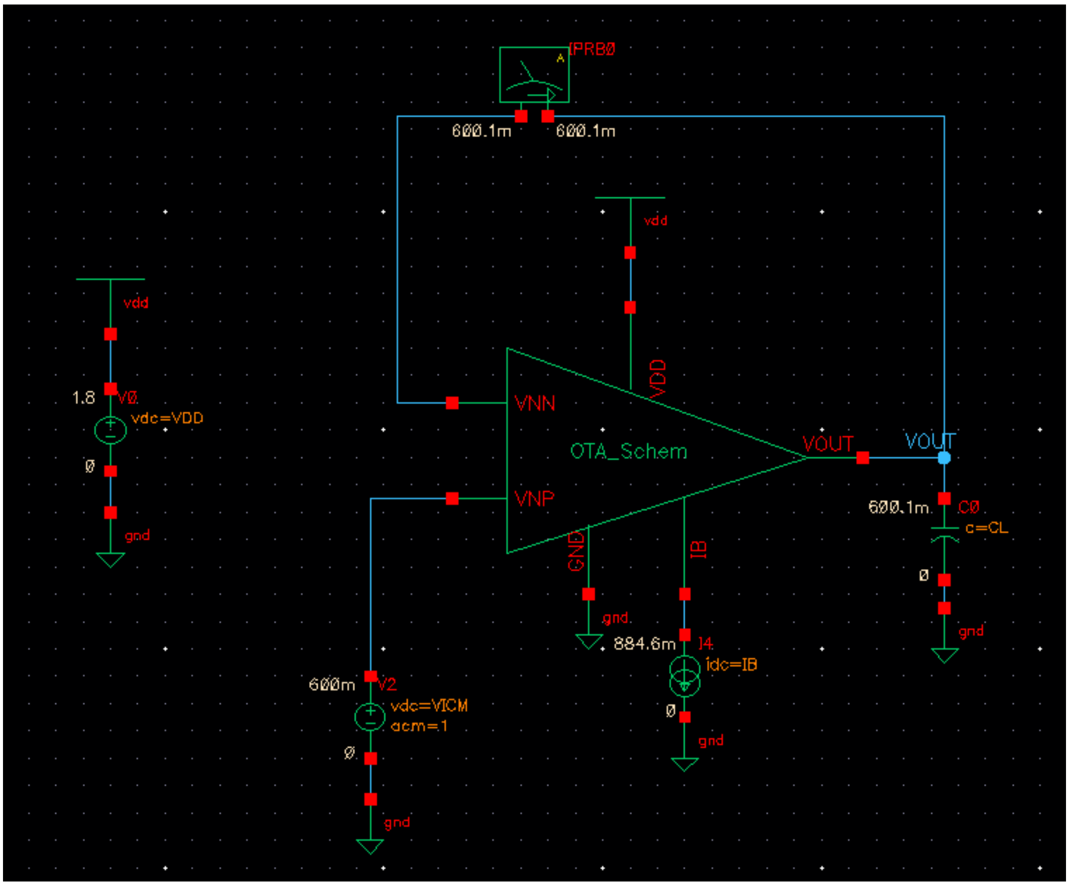


Figure 45 DC Voltages Annotated in TB Schematic

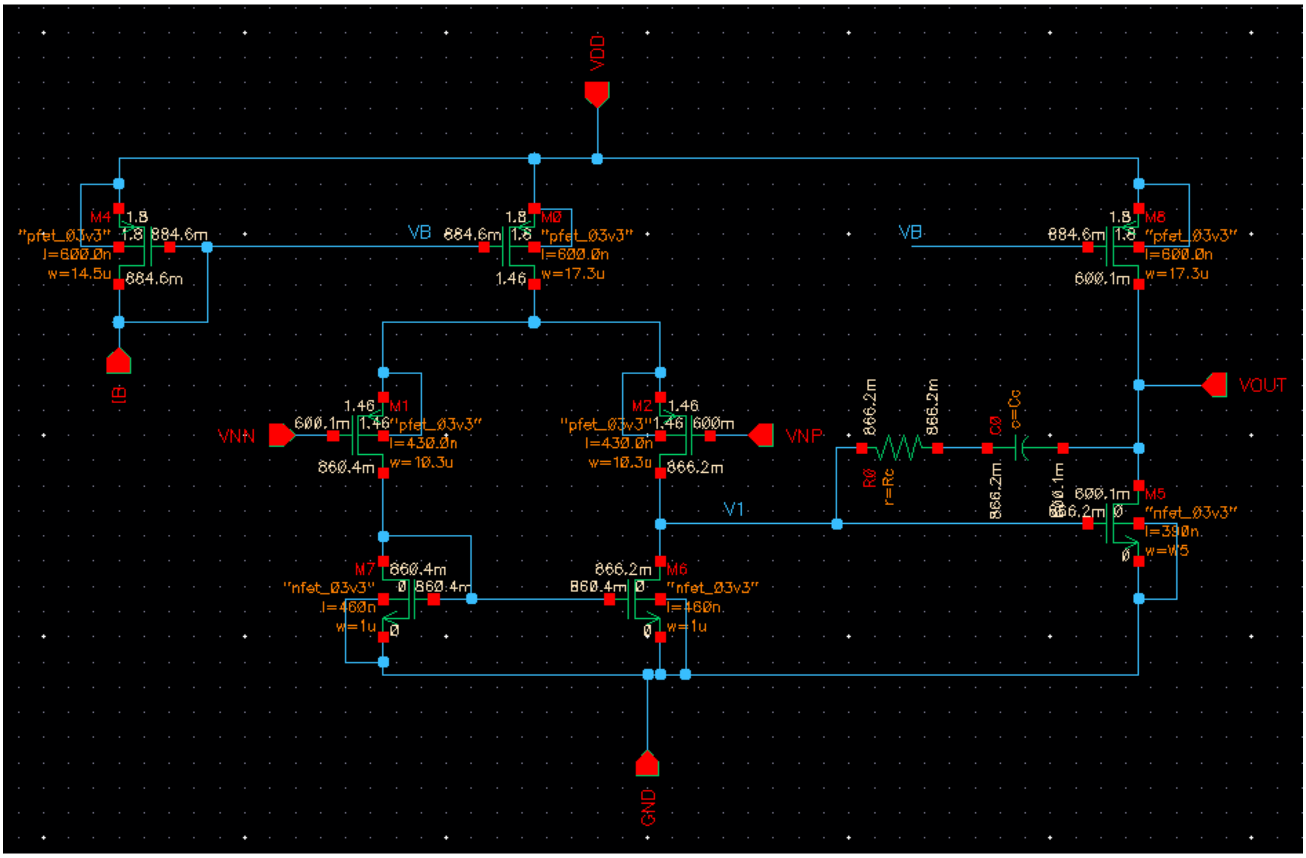


Figure 46 Dc Voltages Annotated in OTA

• Are the DC voltages at the input terminals of the op-amp exactly equal? Why?

There is only a 0.1mV difference between them so approximately equal this is due to the very high gain of the amplifier minimizing Verror.

• Is the DC voltage at the output of the first stage exactly equal to the value in the open-loop simulation? Why?

Its value is 866mV now as opposed to 864mV during the open loop simulation, very small difference between the two due to the high open loop gain

• Is the current (and gm) in the input pair exactly equal? Why?

There is a very small difference that is almost negligible between them, this is due to the very high gain of the amplifier making the input and output nodes follow each other and reduces the deviation caused significantly thus all the values are almost equal to each other.

Loop Gain:

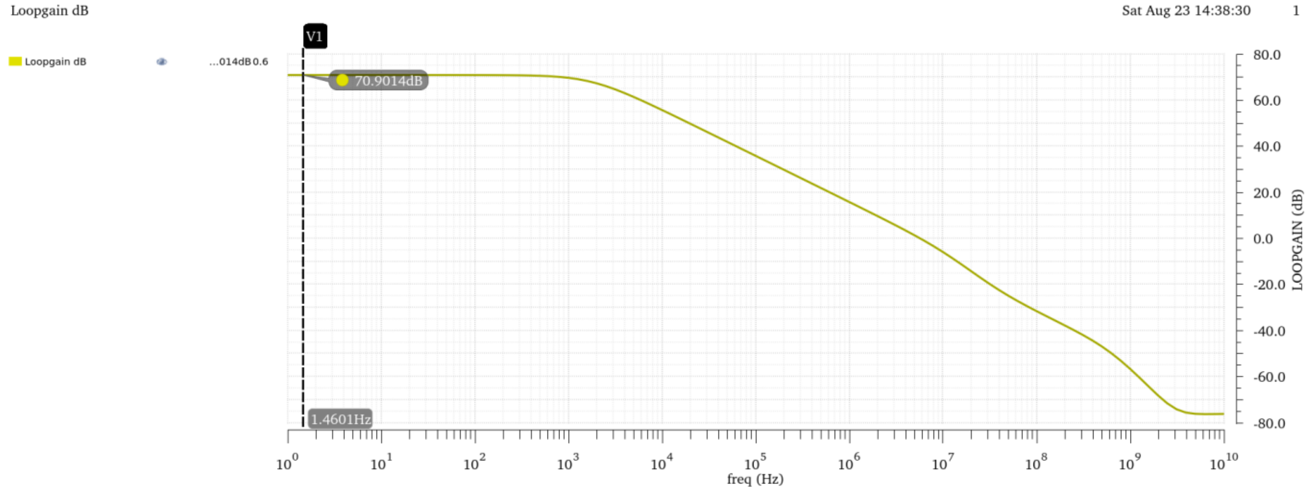


Figure 47 Loop Gain vs Frequency (dB)

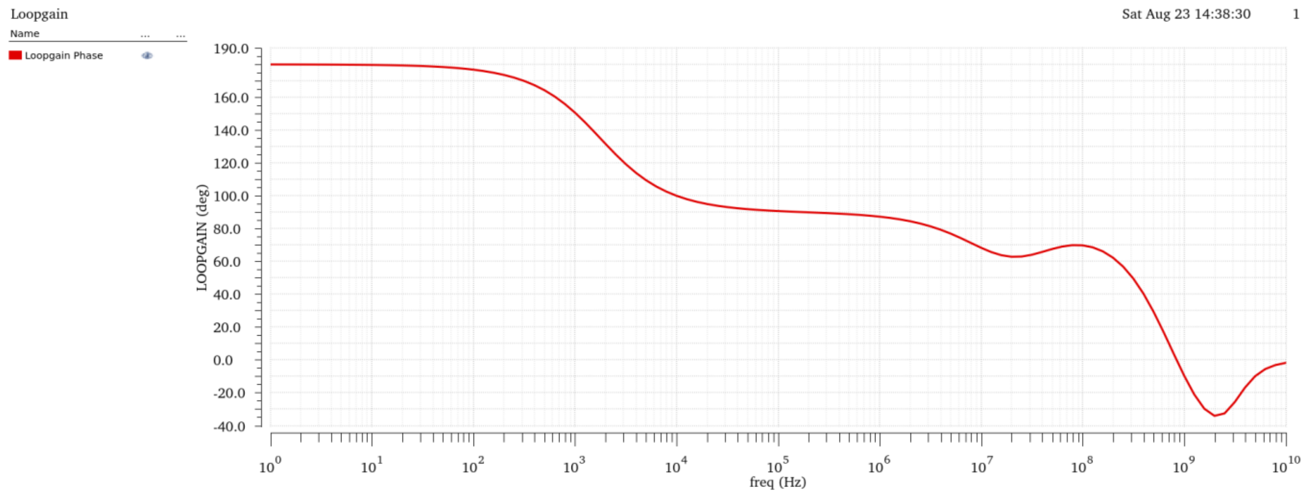


Figure 48 Loop Gain Phase vs Frequency

Compare DC gain, fu, and GBW with those obtained from open-loop simulation. Comment

ymax(mag(getData("loopGain" ?...	3.508k
unityGainFreq(mag(getData("lo...	5.828M
gainBwProd(mag(getData("loop...	6.242M

Figure 49 DC LG Gain, UGF and GBW

All Values are comparable to the Open Loop Simulation values at beta = 1 !
As Loop Gain follows the open loop behaviour.

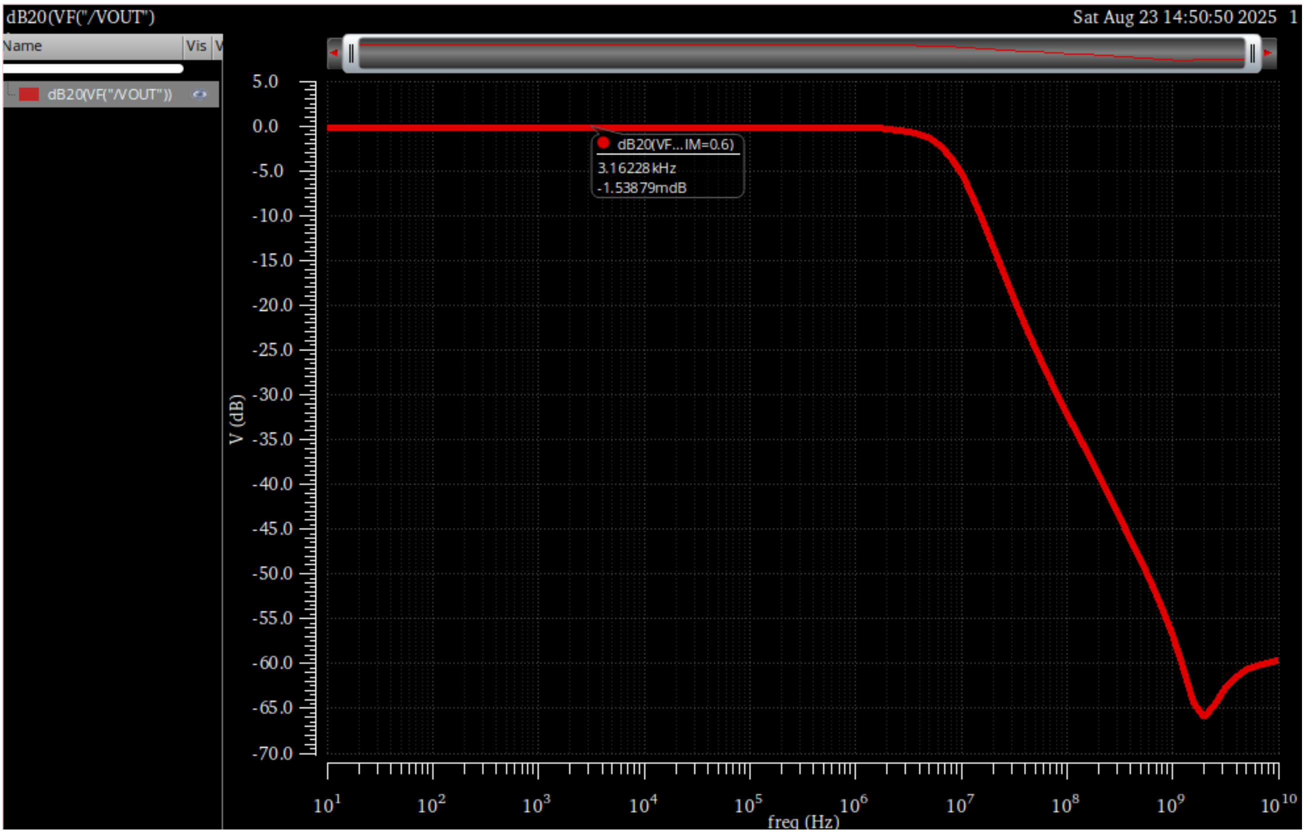


Figure 50 Closed Loop Bandwidth

$$BW_{CL} \approx \mathbf{7.447MHz}$$

Hand Analysis:

$$\beta = 1 \quad , \quad A_{OL} = 3.5\text{K}\,(From\ Open\ Loop\ Simulation)$$
$$Loop\ Gain = \beta A_{OL} = \mathbf{3.5K} \rightarrow 70.88\ dB$$
$$A_{CL} = \frac{A_{OL}}{1 + \beta A_{OL}} = \mathbf{0.99971} \rightarrow \mathbf{-2.48 * 10^{-3}}\ \boldsymbol{dB}$$

	Simulation	Hand Analysis
Beta	1	1
Loop Gain	3.508k	3.5k
Loop Gain (dB)	3.508k	3.5k
Closed Loop Gain	0.9998	0.9997

Analytic Results agree with Hand Analysis for the most part.

Phase Margin:

I had to fine tune the values of Rz and Cc further to get better slew rate and phase margin

Phase Margin	75.41

Figure 51 Phase margin from simulation

$$\boldsymbol{R_z = 5K\Omega \quad , C_C = 2pF}$$

Hand Analysis:

$$PM = \tan^{-1}\left(\frac{\omega_{p2}}{\omega_{gc}}\right) \approx \tan^{-1}\left(\frac{gm_2 C_C}{gm_1 C_L}\right) \approx 70^o$$

	Simulation	Hand Analysis
PM	75.41	70

Hand Analysis isn't entirely accurate because it doesn't account for parasitic and MOSFET capacitances in the circuit.

Slew Rate:

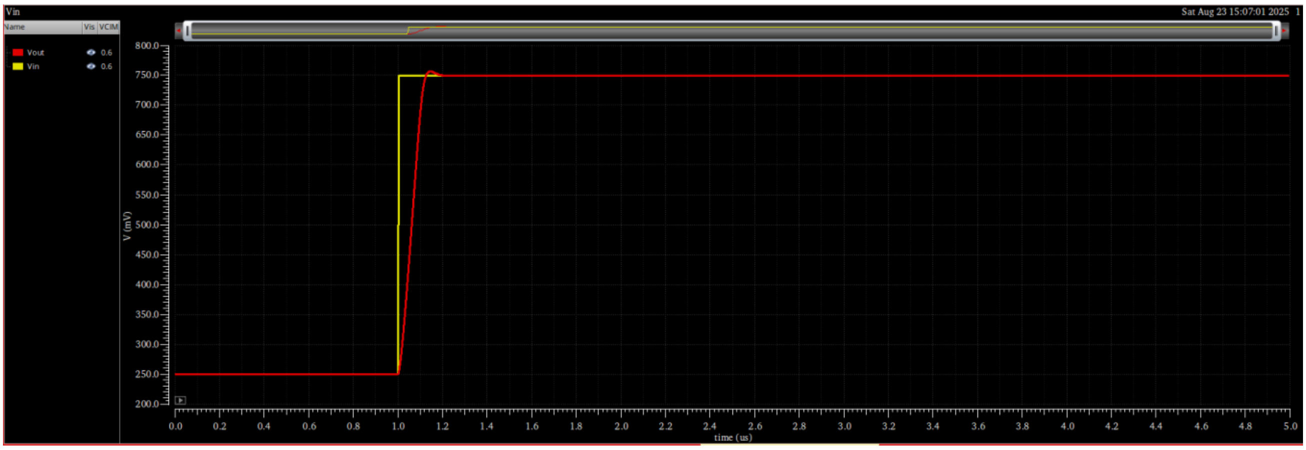


Figure 52 VIn and Vout overlaid

Slewrate	5.138M

Figure 53 Slew Rate from simulation

Hand Analysis:

$$SR = \frac{I_{B1}}{C_C} = \frac{11.53\mu}{2p} = \mathbf{5.765M}$$

	Simulation	Hand Analysis
SR	5.138M	5.765M

Hand Analysis is slightly larger because it doesn't account for extra mosfet and parasitic capacitances in the circuit.

But both are comparable and meet the required spec.

Settling Time:

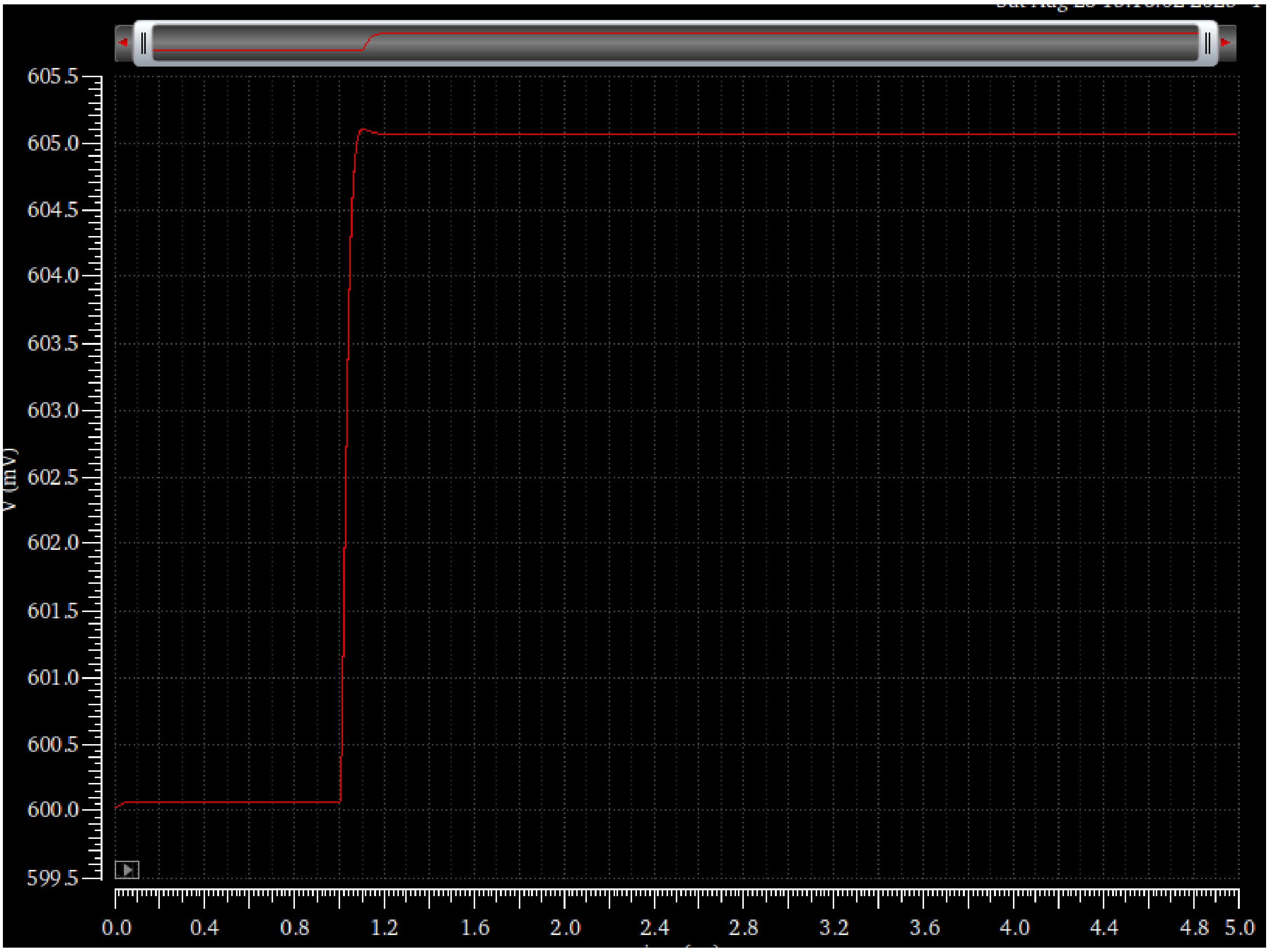


Figure 54 VOUT

riseTime(VT("/VOUT") 0.6 nil 0.6...	44.44n

Figure 55 Simulation Rise Time

Hand Analysis:

$$Rise\ Time = \frac{2.2}{\omega_u} \approx \mathbf{37.9n}$$

	Simulation	Hand Analysis
SR	44.44n	37.9n

both are comparable and meet the required spec.

There is some ringing observed in the output, due to PM being less than 76.3°